

9

METRIC MEASUREMENT

Figure 9.1 A road sign in Finland, a country that uses the metric system. (credit: modification of work by Anna Järvenpää/Flickr, Public Domain)

Chapter Outline

- 9.1 The Metric System
- 9.2 Measuring Area
- 9.3 Measuring Volume
- 9.4 Measuring Weight
- 9.5 Measuring Temperature



Introduction

You are planning a road trip from your home state to the sunny beaches of Mexico and need to prepare a budget. While in the United States, gasoline is sold in gallons and distances are measured in miles, but in almost any other country you will find that gasoline is sold in liters and distance is measured in kilometers.

Whether you're traveling, baking, watching an international sporting event, working with machine tools, or using scientific equipment, it's important to understand the **metric system**, or the International System of Units (SI). The metric system is a decimal measuring system that uses meters, liters, and grams to quantify length, capacity, and mass. It is used in all but three countries in the world, including the United States.

9.1 The Metric System



Figure 9.2 A scale that measures weight in both metric and customary units. (credit: “Weighing the homemade cheddar” by Ruth Hartnup/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Identify units of measurement in the metric system and their uses.
2. Order the six common prefixes of the metric system.
3. Convert between like unit values.

Even if you don't travel outside of the United States, many specialty grocery stores utilize the metric system. For example, if you want to make authentic tamales you might visit the nearest Hispanic grocery store. While shopping, you discover that there are two brands of masa for the same price, but one bag is marked 1,200 g and the other 1 kg. Which one is the better deal? Understanding the metric system allows you to understand that 1,200 grams is equivalent to 1.2 kg, so the 1,200 g bag is the better deal.

Units of Measurement in the Metric System

Units of measurement provide common standards so that regardless of where or when an object or substance is measured, the results are consistent. When measuring distance, the units of measure might be feet, **meters**, or miles. Weight might be expressed in terms of pounds or **grams**. Volume or capacity might be measured in gallons, or **liters**. Understanding how metric units of measure relate to each other is essential to understanding the metric system:

- The metric unit for distance is the meter (m). A person's height might be written as 1.8 meters (1.8 m). A meter slightly longer than a yard (3 feet), while a centimeter is slightly less than half an inch.
- The metric unit for area is the **square meter** (m^2). The area of a professional soccer field is 7,140 square meters ($7,140 \text{ m}^2$).
- The metric base unit for volume is the **cubic meter** (m^3). However, the liter (L), which is a metric unit of capacity, is used to describe the volume of liquids. Soda is often sold in 2-liter (2 L) bottles.
- The gram (g) is a metric unit of mass but is commonly used to express weight. The weight of a paper clip is approximately 1 gram (1 g).
- The metric unit for temperature is **degrees Celsius** ($^{\circ}\text{C}$). The temperature on a warm summer day might be 24°C .

While the U.S. Customary System of Measurement uses ounces and pounds to distinguish between weight units of different sizes, in the metric system a *base unit* is combined with a prefix, such as *kilo-* in kilogram, to identify the relationship between smaller or larger units.

 *When using abbreviations to represent metric measures, always separate the quantity and the units with a space, with no spaces between the letters or symbols in the units. For example, 7 millimeters is written as 7 mm, not 7mm.*

TECH CHECK

It is important to be able to convert between the U.S. Customary System of Measurement and the metric system. However, in this chapter we'll focus on converting units within the metric system. Why? Typing "200 centimeters in inches" into any browser search bar will instantly convert those measures for you (Figure 9.3). You'll have an opportunity in the [Projects](#) to work between measurement systems.

The screenshot shows a Google search bar with the text "200 centimeters to inches". Below the search bar, the results show "About 16,500,000 results (0.47 seconds)". A conversion tool is displayed, showing "200" in the "Length" field, "Centimeter" in the unit dropdown, and "78.7402" in the "Inch" field. Below the tool, a formula is shown: "Formula divide the length value by 2.54".

Figure 9.3 (credit: Screenshot/Google)

Let's be honest. Most of us use computers or smartphones to perform many of the calculations and conversions we were taught in math class. But there is value in understanding the metric system since it exists all around us, and most importantly, knowing how the different metric units relate to each other allows you to compare prices, find the right tool in a workshop, or acclimate when in another country. No matter the circumstance, you cannot avoid the metric system.

EXAMPLE 9.1

Determining the Correct Base Unit

Which base unit would be used to express the following?

1. amount of water in a swimming pool
2. length of an electrical wire
3. weight of one serving of peanuts

✓ Solution

1. Liquid volume is generally expressed in units of liters (L).
2. Length is measured in units of meters (m).
3. Weight is commonly expressed in units of grams (g).

> YOUR TURN 9.1

Determine the correct base measurement for each of the following.

1. weight of a laptop
2. width of a table
3. amount of soda in a pitcher

While there are other base units in the metric system, our discussions in this chapter will be limited to units used to express length, area, volume, weight, and temperature.

Metric Prefixes

Unlike the U.S. Customary System of Measurement in which 12 inches is equal to 1 foot and 3 feet are equal to 1 yard, the metric system is structured so that the units within the system get larger or smaller by a power of 10. For example, a centimeter is 10^{-2} , or 100 times smaller than a meter, while the kilometer is 10^3 , or 1,000 times larger than a meter.

The metric system combines base units and unit prefixes reasonable to the size of a measured object or substance. The most used prefixes are shown in Table 9.1. An easy way to remember the order of the prefixes, from largest to smallest, is the mnemonic *King Henry Died From Drinking Chocolate Milk*.

Prefix	kilo-	hecto-	deca-	base unit	deci-	cent-	milli-
Abbreviation	k	h	da		d	c	m
Magnitude	10^3	10^2	10^1	10^0 or 1	10^{-1}	10^{-2}	10^{-3}

Table 9.1 Metric Prefixes

EXAMPLE 9.2

Ordering the Magnitude of Units

Order the measures from smallest unit to largest unit.

centimeter, millimeter, decimeter

Solution

Looking at the metric prefixes, we can see that the prefix order from smallest unit to largest unit is milli-, centi-, deci-, so the order of the units from smallest to largest is millimeter, centimeter, decimeter.

YOUR TURN 9.2

1. Order the measures from largest unit to smallest unit.
hectogram, decagram, kilogram

EXAMPLE 9.3

Determining Reasonable Values for Length

What is a reasonable value for the length of a person's thumb: 5 meters, 5 centimeters, or 5 millimeters?

Solution

Given that a meter is slightly longer than a yard, 5 meters is not a reasonable value for the length of a person's thumb. Since a millimeter is 10 times smaller than a centimeter, which is approximately $\frac{1}{2}$ inch, 5 millimeters is not a reasonable estimate for the length of a person's thumb. The correct answer is 5 centimeters.

YOUR TURN 9.3

1. What is a reasonable estimate for the length of a hallway: 2.5 kilometers, 2.5 meters, or 2.5 centimeters?

Converting Metric Units of Measure

Imagine you order a textbook online and the shipping detail indicates the weight of the book is 1 kg. By attaching the letter “k” to the base unit of gram (g), the unit used to express the measure is 10^3 or 1,000 times greater than a gram. One kilogram is equivalent to 1,000 grams.

The tip of a highlighter measures approximately 1 cm. The letter “c” attached to the base unit of meter (m) means the

unit used to express the measure is 10^{-2} or $\frac{1}{100}$ of a meter. One meter is equivalent to 100 centimeters.

A **conversion factor** is used to convert from smaller metric units to bigger metric units and vice versa. It is a number that when used with multiplication or division converts from one metric unit to another, both having the same base unit. In the metric system, these conversion factors are directly related to the powers of 10. The most common used conversion factors are shown in Figure 9.4.

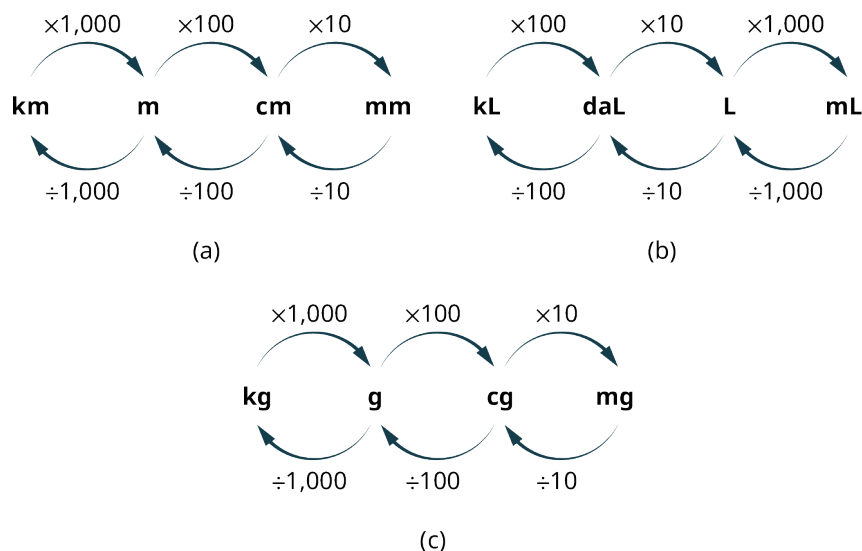


Figure 9.4 Common Metric Conversion Factors for (a) Meters, (b) Liters, and (c) Grams

EXAMPLE 9.4

Converting Metric Distances Using Multiplication

The firehouse is 13.45 km from the library. How many meters is it from the firehouse to the library?

Solution

When converting from a larger unit to a smaller unit, use multiplication. The conversion factor from kilometer to the base unit of meters is 1,000.

$$13.45 \times 1,000 = 13,450$$

So, the firehouse is 13,450 meters away from the firehouse.

YOUR TURN 9.4

1. The record for the men's high jump is 2.45 m. What is the record when expressed in centimeters?

EXAMPLE 9.5

Converting Metric Capacity Using Division

How many liters is 3,565 milliliters?

Solution

When converting from a smaller unit to a larger unit, use division. The conversion factor from milliliter to the base unit of liters is 1,000.

$$3,565 \div 1,000 = 3.565$$

So, 3,565 milliliters is 3.565 liters.

> YOUR TURN 9.5

1. A bottle of cleaning solution measures 7.6 liters. How many decaliters is that?

EXAMPLE 9.6**Converting Metric Units of Mass to Solve Problems**

Caroline and Aiyana are working on a chemistry experiment together and must perform calculations using measurements taken during the experiment. Due to miscommunication, Caroline took measurements in centigrams and Aiyana used milligrams. Convert Caroline's measurement of 125 centigrams to milligrams.

✓ Solution

When converting from a larger unit to a smaller unit, use multiplication. The conversion factor from centigrams to the milligrams is 10.

$$125 \times 10 = 1,250$$

So, the 125 centigrams are 1,250 milligrams.

> YOUR TURN 9.6

1. Convert Aiyana's measurement of 1,457 mg to centigrams.

EXAMPLE 9.7**Converting Metric Units of Volume to Solve Problems**

A bottle contains 500 mL of juice. If the juice is packaged in 24-bottle cases, how many liters of juice does the case contain?

✓ Solution

Step 1: Multiply the amount of juice in each bottle by the number of bottles.

$$500 \text{ mL} \times 24 = 12,000 \text{ mL}$$

Step 2: Divide by 1,000 to convert from milliliters to liters.

$$\frac{12,000 \text{ mL}}{1,000} = 12 \text{ L}$$

So, there are 12 liters of juice in each case.

> YOUR TURN 9.7

1. A hospital orders 250 doses of liquid amoxicillin. Each dose is 5 mL. How many liters of amoxicillin did the hospital order?

**PEOPLE IN MATHEMATICS**

Valerie Antoine

In the 1970s, people were told that they must learn the metric system because the United States was soon going to

convert to using metric measurements. Children and young adults probably watched educational cartoons about the metric system on Saturday mornings.

In 1975, President Gerald Ford signed the Metric Conversion Act and created a board of 17 people commissioned to coordinate the voluntary switch to the metric system in the United States. Among those 17 people was Valerie Antoine, an engineer who made it her life's work to push for this change. Despite President Ronald Reagan dissolving the board in 1982, effectively killing the move to the metric system at the time, Antoine continued the movement out of her own home as the executive director of the U.S. Metric Association. Reagan's decision followed intensive lobbying by American businesses whose factories used machinery designed to use customary measurements by workers trained in customary measurements. There was also intense public pressure from American citizens who didn't want to go through the time consuming and expensive process of changing the country's entire infrastructure. Fueled by a Congressional mandate in 1992 that required all federal agencies make the switch to the metric system, Antoine never gave up hope that the metric system would trickle down from the government and find its way into American schools, homes, and everyday life.

▶ VIDEO

[U.S. Office of Education: Metric Education \(https://openstax.org/r/U.S._Office_Education\)](https://openstax.org/r/U.S._Office_Education)

EXAMPLE 9.8

Converting Grams to Solve Problems

The nutrition label on a jar of spaghetti sauce indicates that one serving contains 410 mg of sodium. You have poured two servings over your favorite pasta before recalling your doctor's advice about keeping your sodium consumption below 1 g per meal. Have you followed your doctor's recommendation?

✓ Solution

Step 1: Multiply the number of servings by the amount of sodium in each serving.

$$410 \text{ mg} \times 2 = 820 \text{ mg}$$

Step 2: Divide by 1,000 to convert from milligrams to grams.

$$\frac{820 \text{ mg}}{1,000} = 0.82 \text{ g}$$

You have followed doctor's recommendation because 0.82 g is less than 1 gram.

> YOUR TURN 9.8

1. The FDA recommends that you consume less than 0.5 g of caffeine daily. A cup of coffee contains 95 mg of caffeine and a can of soda contains 54 mg. If you drink 2 cups of coffee and 3 cans of soda, have you kept your day's caffeine consumption to the FDA recommendation? Explain.

EXAMPLE 9.9

Comparing Different Units

A student carefully measured 0.52 cg of copper for a science experiment, but their lab partner said they need 6 mg of copper total. How many more centigrams of copper does the student need to add?

✓ Solution

Step 1: Convert these two measurements into a common unit. Since the question asks for the number of centigrams,

convert 6 mg to centigrams, which is 0.6 cg.

Step 2: Find the difference by subtracting $0.6 - 0.52$ which is 0.08 cg. This means the student must add another 0.08 cg of copper.

YOUR TURN 9.9

1. Kyrie boasted he jumped out of an airplane at an altitude of 3,810 meters on his latest skydive trip. His friend said they beat Kyrie because their jump was at an altitude of 3.2 km. Whose skydive was at a greater altitude?

WHO KNEW?

The United States and the Metric System

Did you know that the metric system pervades daily life in the United States already? While Americans still may purchase gallons of milk and measure house sizes in square feet, there are many instances of the metric system. Photographers buy 35 mm film and use 50 mm lenses. When you have a headache, you might take 600 mg of ibuprofen. And if you are eating a low-carb diet you probably restrict your carb intake to fewer than 20 g of carbs daily. Did you know even the dollar is metric? In the video, Neil DeGrasse Tyson and comedian co-host Chuck Nice provide an amusing perspective on the metric system.

The International System of Units (SI) is the current international standard metric system and is the most widely used system around the world. In most English-speaking countries SI units such as meter, liter, and metric ton are spelled metre, litre, and tonne.

VIDEO

[Neil deGrasse Tyson Explains the Metric System \(https://openstax.org/r/Tyson_Explains_Metric_System\)](https://openstax.org/r/Tyson_Explains_Metric_System)

WORK IT OUT

Get to Know the Metric System

Just how much is the metric system a part of your life now? Probably more than you think. For the next 24 hours, take notice as you move through your daily activities. When you are shopping, are the package sizes provided in metric units? Change the weather app on your phone to display the temperature in degrees Celsius. Are you able to tell what kind of day it will be now? While the United States is not officially using the metric system, you will still find the metric system all around you.

Check Your Understanding

1. Which metric base unit would be used to measure the height of a door?
2. Which metric base unit would be used to measure your weight?
3. Which is greater: 12 hectoliters or 12 centiliters?
4. Convert 1,520 cm to meters (m).
5. Convert 1.34 km to decameters (dam).
6. Convert 12,700 cg to hectograms (hg).
7. Convert 750 km to millimeters (mm).
8. Which is the larger measurement: 0.04 dam or 40 cm?



SECTION 9.1 EXERCISES

For the following exercises, determine the base unit of the metric system described. Choose from *liter*, *gram*, or *meter*.

1. amount of soda in a bottle
2. weight of a book you are mailing
3. amount of gasoline needed to fill a car's tank
4. height of a computer desk
5. weight of a dog at the veterinarian's office
6. dimensions of the newest HD TV
7. distance a student athlete ran on the treadmill during their latest workout
8. amount of water to add to bleach for mopping floors
9. Write the order of the metric prefixes from greatest to least.

For the following exercises, choose the smaller of the two units.

10. decagrams or decigrams
11. centimeters or millimeters
12. liters or kiloliters
13. decigrams or centigrams
14. decameters or hectometers
15. milliliters or deciliters
16. Convert 158 hectometers to meters (m).
17. Convert 12.3 cg to grams (g).
18. Convert 160 dam to kilometer (km).
19. You purchase 10 kg of candy. You divide the candy into 25 bags. How many grams of candy are in each bag?
20. An outdoor track is 400 m long. If you run 10 laps around the track, how many kilometers have you run?
21. An aspirin tablet is 650 mg. If you take 2 aspirin twice in one day, how many grams of aspirin have you taken?
22. A juice box contains 450 mL of juice. If there are 6 juice boxes in a package, how many liters of juice are in the package?
23. Carlos consumed 4 cans of his favorite energy drink. If each can contains 111 mg of caffeine, how many grams of caffeine did he consume?
24. Celeste ran a total distance of 72,548 m in 1 week while training for a 5K fun run. How many kilometers did she run?
25. During week one of their diet, Dakota consumed 413 g of carbs. After speaking to their doctor, they only consumed 210 g of carbs the second week. How many fewer milligrams of carbs did they consume the second week?
26. Kaylea gained 2.3 kg of muscle weight in 9 months of working out. Cho gained 250 decagrams of muscle during that same time. Who gained more muscle weight? How many grams more?

9.2 Measuring Area



Figure 9.5 A painter uses an extension roller to paint a wall. (credit: "Paint Rollers are effective" by WILLPOWER STUDIOS/ Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Identify reasonable values for area applications.
2. Convert units of measures of area.
3. Solve application problems involving area.

Area is the size of a surface. It could be a piece of land, a rug, a wall, or any other two-dimensional surface with attributes that can be measured in the metric unit for distance-meters. Determining the area of a surface is important to many everyday activities. For example, when purchasing paint, you'll need to know how many **square units** of surface area need to be painted to determine how much paint to buy.

Square units indicate that two measures in the same units have been multiplied together. For example, to find the area of a rectangle, multiply the length units and the width units to determine the area in square units.

$$1 \text{ cm} \times 1 \text{ cm} = 1 \text{ cm}^2$$

Note that to accurately calculate area, each of the measures being multiplied must be of the same units. For example, to find an area in square centimeters, both length measures (length and width) must be in centimeters.

FORMULA

The formula used to determine area depends on the shape of that surface. Here we will limit our discussions to the area of rectangular-shaped objects like the one in Figure 9.6. Given this limitation, the basic formula for area is:

$$\text{Area} = \text{length } (l) \times \text{width } (w)$$

or

$$A = l \times w$$

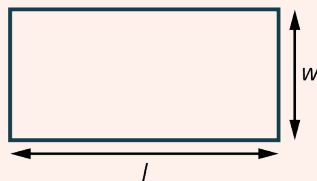


Figure 9.6 Rectangle with Length (l) and Width (w) Labeled

Reasonable Values for Area

Because area is determined by multiplying two lengths, the magnitude of difference between different square units is exponential. In other words, while a meter is 100 times greater in length than a centimeter, a square meter (m^2) is 100×100 , or 10,000 times greater in area than a square centimeter (cm^2). The relationships between benchmark metric area units are shown in the following table.

Units	Relationship	Conversion Rate
km^2 to m^2	$\text{km} \times \text{km} = \text{km}^2$ $1 \text{ km} = 1,000 \text{ m}$ $1,000 \text{ m} \times 1,000 \text{ m} = 1,000,000 \text{ m}^2$	$1 \text{ km}^2 = 1,000,000 \text{ m}^2$
m^2 to cm^2	$\text{m} \times \text{m} = \text{m}^2$ $1 \text{ m} = 100 \text{ cm}$ $100 \text{ cm} \times 100 \text{ cm} = 10,000 \text{ cm}^2$	$1 \text{ m}^2 = 10,000 \text{ cm}^2$
cm^2 to mm^2	$\text{cm} \times \text{cm} = \text{cm}^2$ $1 \text{ cm} = 10 \text{ mm}$ $10 \text{ mm} \times 10 \text{ mm} = 100 \text{ mm}^2$	$1 \text{ cm}^2 = 100 \text{ mm}^2$

An essential understanding of metric area is to identify reasonable values for area. When testing for reasonableness you should assess both the unit and the unit value. Only by examining both can you determine whether the given area is reasonable for the situation.

EXAMPLE 9.10

Determining Reasonable Units for Area

Which unit of measure is most reasonable to describe the area of a sheet of paper: km^2 , cm^2 , or mm^2 ?

Solution

In the U.S. Customary System of Measurement, the length and width of paper is usually measured in inches. In the metric system centimeters are used for measures usually expressed in inches. Thus, the most reasonable unit of measure to describe the area of a sheet of paper is square centimeters. Square kilometers is too large a unit and square millimeters is too small a unit.

YOUR TURN 9.10

1. Which unit of measure is most reasonable to describe the area of a forest: km^2 , cm^2 , or mm^2 ?

EXAMPLE 9.11

Determining Reasonable Values for Area

You want to paint your bedroom walls. Which represents a reasonable value for the area of the walls: 100 cm^2 , 100 m^2 , or 100 km^2 ?

Solution

An area of 100 cm^2 is equivalent to a surface of $10 \text{ cm} \times 10 \text{ cm}$, which is much too small for the walls of a bedroom. An area of 100 km^2 is equivalent to a surface of $10 \text{ km} \times 10 \text{ km}$, which is much too large for the walls of a bedroom. So, a reasonable value for the area of the walls is 100 m^2 .

YOUR TURN 9.11

1. Which represents a reasonable value for the area of the top of a kitchen table: $1,800 \text{ mm}^2$, $1,800 \text{ cm}^2$, or $1,800 \text{ m}^2$?

EXAMPLE 9.12**Explaining Reasonable Values for Area**

A landscaper is hired to resod a school's football field. After measuring the length and width of the field they determine that the area of the football field is $5,350 \text{ km}^2$. Does their calculation make sense? Explain your answer.

Solution

No, kilometers are used to determine longer distances, such as the distance between two points when driving. A football field is less than 1 kilometer long, so a more reasonable unit of value would be m^2 . An area of $5,350 \text{ km}^2$ can be calculated using the dimensions 53.5 by 100, which are reasonable dimensions for the length and width of a football field. So, a more reasonable value for the area of the football field is $5,350 \text{ m}^2$.

YOUR TURN 9.12

1. A crafter uses four letter-size sheets of paper to create a paper mosaic in the shape of a square. They decide they want to frame the paper mosaic, so they measure and determine that the area of the paper mosaic is $2,412.89 \text{ cm}^2$. Does their calculation make sense? Explain your answer.

Converting Units of Measures for Area

Just like converting units of measure for distance, you can convert units of measure for area. However, the conversion factor, or the number used to multiply or divide to convert from one area unit to another, is not the same as the conversion factor for metric distance units. Recall that the conversion factor for area is exponentially relative to the conversion factor for distance. The most frequently used conversion factors are shown in [Figure 9.7](#).

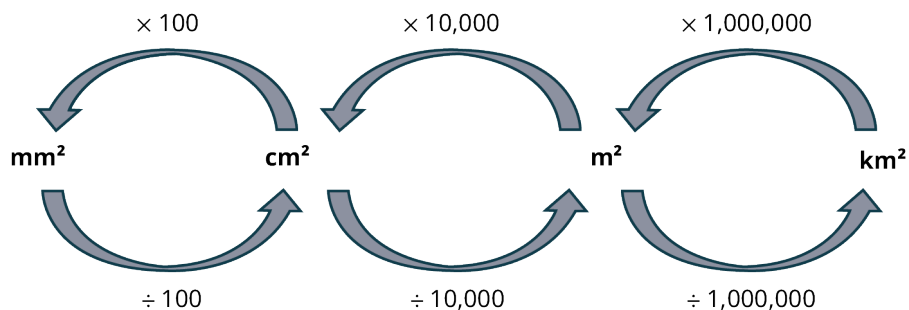


Figure 9.7 Common Conversion Factors for Metric Area Units

VIDEO

[Converting Metric Units of Area \(https://openstax.org/r/Metric_Units_of_Area\)](https://openstax.org/r/Metric_Units_of_Area)

EXAMPLE 9.13**Converting Units of Measure for Area Using Division**

A plot of land has an area of $237,500,000 \text{ m}^2$. What is the area in square kilometers?

Solution

Use division to convert from a smaller metric area unit to a larger metric area unit. To convert from m^2 to km^2 , divide the value of the area by 1,000,000.

$$\frac{237,500,000}{1,000,000} = 237.5$$

The plot of land has an area of 237.5 km².

YOUR TURN 9.13

1. A roll of butcher paper has an area of 1,532,900 cm². What is the area of the butcher paper in square meters?

EXAMPLE 9.14

Converting Units of Measure for Area Using Multiplication

A plot of land has an area of 0.004046 km². What is the area of the land in square meters?

Solution

Use multiplication to convert from a larger metric area unit to a smaller metric area unit. To convert from km² to m², multiply the value of the area by 1,000,000.

$$0.004046 \times 1,000,000 = 4,046$$

The plot of land has an area of 4,046 m².

YOUR TURN 9.14

1. A bolt of fabric has an area of 136.5 m². What is the area of the bolt of fabric in square centimeters?

EXAMPLE 9.15

Determining Area by Converting Units of Measure for Length First

A computer chip measures 10 mm by 15 mm. How many square centimeters is the computer chip?

Solution

Step 1: Convert the measures of the computer chip into centimeters

$$10 \text{ mm} = 1 \text{ cm}$$

$$15 \text{ mm} = 1.5 \text{ cm}$$

Step 2: Use the [area formula](#) to determine the area of the chip.

$$1 \times 1.5 = 1.5$$

The computer chip has an area of 1.5 cm².

YOUR TURN 9.15

1. A piece of fabric measures 100 cm by 106 cm. What is the area of the fabric in square meters?

Solving Application Problems Involving Area

While it may seem that solving area problems is as simple as multiplying two numbers, often determining area requires more complex calculations. For example, when measuring the area of surfaces, you may need to account for portions of the surface that are not relevant to your calculation.

EXAMPLE 9.16**Solving for the Area of Complex Surfaces**

One side of a commercial building is 12 meters long by 9 meters high. There is a rolling door on this side of the building that is 4 meters wide by 3 meters high. You want to refinish the side of the building, but not the door, with aluminum siding. How many square meters of aluminum siding are required to cover this side of the building?

 Solution

Step 1: Determine the area of the side of the building.

$$12 \text{ m} \times 9 \text{ m} = 106 \text{ m}^2$$

Step 2: Determine the area of the door.

$$4 \text{ m} \times 3 \text{ m} = 12 \text{ m}^2$$

Step 3: Subtract the area of the door from the area of the side of the building.

$$106 \text{ m}^2 - 12 \text{ m}^2 = 92 \text{ m}^2$$

So, you need to purchase 92 m^2 of aluminum siding.

**YOUR TURN 9.16**

1. You want to cover a garden with topsoil. The garden is 5 meters by 8 meters. There is a path in the middle of the garden that is 8 meters long and 0.75 meters wide. What is the area of the garden you need to cover with topsoil?

When calculating area, you must ensure that both distance measurements are expressed in terms of the same distance units. Sometimes you must convert one measurement before using the [area formula](#).

EXAMPLE 9.17**Solving for Area with Distance Measurements of Different Units**

A national park has a land area in the shape of a rectangle. The park measures 2.2 kilometers long by 1,250 meters wide. What is the area of the park in square kilometers?

 Solution

Step 1: Use a conversion fraction to convert the information given in meters to kilometers.

$$1,250 \text{ m} \times \frac{1 \text{ km}}{1,000 \text{ m}} = 1.25 \text{ km}$$

Step 2: Multiply to find the area.

$$2.2 \text{ km} \times 1.25 \text{ km} = 2.75 \text{ km}^2$$

The park has an area of 2.75 km^2 .

**YOUR TURN 9.17**

1. An Olympic pool measures 50 meters by 2,500 centimeters. What is the surface area of the pool in square

meters?

When calculating area, you may need to use multiple steps, such as converting units and subtracting areas that are not relevant.

EXAMPLE 9.18**Solving for Area Using Multiple Steps**

A kitchen floor has an area of 15 m^2 . The floor in the kitchen pantry is 100 cm by 200 cm . You want to tile the kitchen and pantry floors using the same tile. How many square meters of tile do you need to buy?

 Solution

Step 1: Determine the area of the pantry floor in square centimeters.

$$100 \text{ cm} \times 200 \text{ cm} = 20,000 \text{ cm}^2$$

Step 2: Divide the area in cm^2 by the conversion factor to determine the area in m^2 since the other measurement for the kitchen floor is in m^2 .

$$\frac{20,000}{10,000} = 2$$

The area of the kitchen pantry floor is 2 m^2 .

Step 3: Add the two areas of the pantry and the kitchen floors together.

$$15 \text{ m}^2 + 2 \text{ m}^2 = 17 \text{ m}^2$$

So, you need to buy 17 m^2 of tile.

**YOUR TURN 9.18**

1. Your bedroom floor has an area of 25 m^2 . The living room floor measures 600 cm by 750 cm . How many square meters of carpet do you need to buy to carpet the floors in both rooms?

 WHO KNEW?**The Origin of the Metric System**

The metric system is the official measurement system for every country in the world except the United States, Liberia, and Myanmar. But did you know it originated in France during the French Revolution in the late 18th century? At the time there were over 250,000 different units of weights and measures in use, often determined by local customs and economies. For example, land was often measured in days, referring to the amount of land a person could work in a day.

**VIDEO**

[Why the Metric System Matters \(https://openstax.org/r/Metric_System_Matters\)](https://openstax.org/r/Metric_System_Matters)

Check Your Understanding

For the following exercises, determine the most reasonable value for each area.

9. bedroom wall: 12 km^2 , 12 m^2 , 12 cm^2 , or 12 mm^2
10. city park: $1,200 \text{ km}^2$, $1,200 \text{ m}^2$, $1,200 \text{ cm}^2$, or $1,200 \text{ mm}^2$

11. kitchen table: 2.5 km^2 , 2.5 m^2 , 2.5 cm^2 , or 2.5 mm^2

For the following exercises, convert the given area to the units shown.

12. $20,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$
 13. $5.7 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$
 14. $217 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ mm}^2$
15. A wall measures 4 m by 2 m. A doorway in the wall measures 0.5 m by 1.6 m. What is the area of the wall not taken by the door in square meters?



SECTION 9.2 EXERCISES

For the following exercises, determine the most reasonable value for each area.

- kitchen floor: 16 km^2 , 16 m^2 , 16 cm^2 , or 16 mm^2
- national park: $1,000 \text{ km}^2$, $1,000 \text{ m}^2$, $1,000 \text{ cm}^2$, or $1,000 \text{ mm}^2$
- classroom table: 5 km^2 , 5 m^2 , 5 cm^2 , or 5 mm^2
- window: $9,000 \text{ km}^2$, $9,000 \text{ m}^2$, $9,000 \text{ cm}^2$, or $9,000 \text{ mm}^2$
- paper napkin: $10,000 \text{ km}^2$, $10,000 \text{ m}^2$, $10,000 \text{ cm}^2$, or $10,000 \text{ mm}^2$
- parking lot: $45,000 \text{ km}^2$, $45,000 \text{ m}^2$, $45,000 \text{ cm}^2$, or $45,000 \text{ mm}^2$

For the following exercises, convert the given area to the units shown.

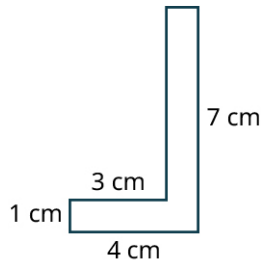
- $700,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$
- $24 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$
- $985 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ mm}^2$
- $4 \text{ km}^2 = \underline{\hspace{2cm}} \text{ m}^2$
- $0.00005 \text{ km}^2 = \underline{\hspace{2cm}} \text{ cm}^2$
- $68,500,000 \text{ mm}^2 = \underline{\hspace{2cm}} \text{ m}^2$
- $0.005 \text{ m}^2 = \underline{\hspace{2cm}} \text{ mm}^2$
- $7,800 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$
- $34.5 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$
- $0.05 \text{ km}^2 = \underline{\hspace{2cm}} \text{ m}^2$
- $0.073 \text{ m}^2 = \underline{\hspace{2cm}} \text{ mm}^2$
- $27,750 \text{ mm}^2 = \underline{\hspace{2cm}} \text{ m}^2$

For the following exercises, determine the area.

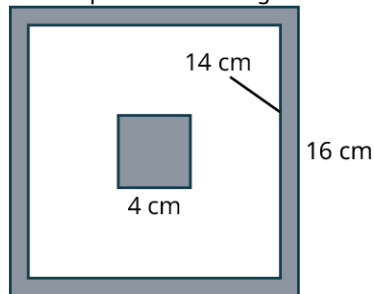
- $l = 300 \text{ cm}$, $w = 4 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ m}^2$
 - $l = 3,500 \text{ mm}$, $w = 0.5 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ cm}^2$
 - $l = 5.5 \text{ km}$, $w = 2,750 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ km}^2$
 - $l = 2,400 \text{ cm}$, $w = 3.2 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ m}^2$
 - $l = 7 \text{ cm}$, $w = 400 \text{ mm}$
 $A = \underline{\hspace{2cm}} \text{ cm}^2$
 - $l = 2,300 \text{ m}$, $w = 4 \text{ km}$
 $A = \underline{\hspace{2cm}} \text{ km}^2$
 - $l = 4.2 \text{ cm}$, $w = 320 \text{ mm}$
 $A = \underline{\hspace{2cm}} \text{ cm}^2$
 - $l = 5,350 \text{ mm}$, $w = 0.5 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ cm}^2$
 - $l = 1.8 \text{ km}$, $w = 2,300 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ km}^2$
28. A notebook is 200 mm by 300 mm. A sticker on the notebook cover measures 2 cm by 2 cm. How many square centimeters of the notebook cover is still visible?
29. A bedroom wall is 4 m by 2.5 m. A window on the wall is 1 m by 2 m. How much wallpaper is needed to cover the wall?
30. A wall is 5 m by 2.5 m. A picture hangs on the wall that is 600 mm by 300 mm. How much of the wall, in m^2 , is not

covered by the picture?

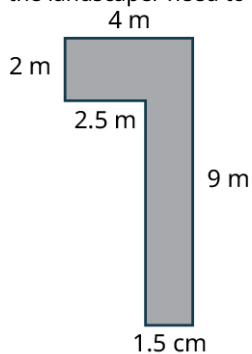
31. What is the area of the shape that is shown?



32. A quilter made a design using a small square, a medium square, and a large square. What is the area of the shaded parts of the design that is shown?



33. A landscaper makes a plan for a walkway in a backyard, as shown. How many square meters of patio brick does the landscaper need to cover the walkway?



34. A painter completed a portrait. The height of the portrait is 36 cm. The width is half as long as the height. What is the area of the portrait in square meters?
35. A room has an area of 137.5 m^2 . The length of the room is 11 m. What is the width of the room?
36. A wall is 4.5 m long by 3 m high. A can of paint will cover an area of 10 m^2 . How many cans of paint are needed if each wall needs 2 coats of paint?
37. A soccer field is 110 meters long and 75 meters wide. If the cost of artificial turf is \$30 per square meter, what is the cost of covering the soccer field with artificial turf?
38. A window is 150 centimeters wide and 90 centimeters high. If three times the area of the window is needed for curtain material, how much curtain material is needed in square meters?
39. A rectangular flower garden is 5.5 meters wide and 8.4 meters long. A path with a width of 1 meter is laid around the garden. What is the area of the path?
40. A dining room is 8 meters wide by 6 meters long. Wood flooring costs \$12.50 per square meter. How much will it cost to install wood flooring in the dining room?

9.3 Measuring Volume



Figure 9.8 Packing cartons sit on a loading dock ready to be filled. (credit: “boxing day” by Erich Ferdinand/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Identify reasonable values for volume applications.
2. Convert between like units of measures of volume.
3. Convert between different unit values.
4. Solve application problems involving volume.

Volume is a measure of the space contained within or occupied by three-dimensional objects. It could be a box, a pool, a storage unit, or any other three-dimensional object with attributes that can be measured in the metric unit for distance—meters. For example, when purchasing an SUV, you may want to compare how many **cubic units** of cargo the SUV can hold.

Cubic units indicate that three measures in the same units have been multiplied together. For example, to find the volume of a rectangular prism, you would multiply the length units by the width units and the height units to determine the volume in square units:

$$1 \text{ cm} \times 1 \text{ cm} \times 1 \text{ cm} = 1 \text{ cm}^3$$

Note that to accurately calculate volume, each of the measures being multiplied must be of the same units. For example, to find a volume in cubic centimeters, each of the measures must be in centimeters.

FORMULA

The formula used to determine volume depends on the shape of the three-dimensional object. Here we will limit our discussions to the area to rectangular prisms like the one in [Figure 9.9](#). Given this limitation, the basic formula for volume is:

$$\text{Volume} = \text{length } (l) \times \text{width } (w) \times \text{height } (h)$$

or

$$V = lwh$$

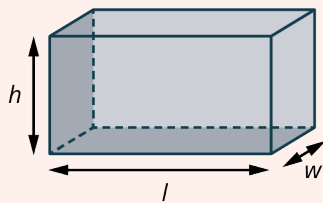


Figure 9.9 Rectangular Prism with Height (*h*), Length (*l*), and Width (*w*) Labeled.

Reasonable Values for Volume

Because volume is determined by multiplying three lengths, the magnitude of difference between different cubic units is exponential. In other words, while a meter is 100 times greater in length than a centimeter, a cubic meter, m^3 , is $100 \times 100 \times 100$, or 1,000,000 times greater in area than a cubic centimeter, cm^3 . This relationship between benchmark metric volume units is shown in the following table.

Units	Relationship	Conversion Rate
km^3 to m^3	$\text{km} \times \text{km} \times \text{km} = \text{km}^3$ $1 \text{ km} = 1,000 \text{ m}$ $1,000 \text{ m} \times 1,000 \text{ m} \times 1,000 \text{ m} = 1,000,000,000 \text{ m}^3$	$1 \text{ km}^3 = 1,000,000,000 \text{ m}^3$
m^3 to dm^3	$\text{m} \times \text{m} \times \text{m} = \text{m}^3$ $1 \text{ m} = 10 \text{ dm}$ $10 \text{ dm} \times 10 \text{ dm} \times 10 \text{ dm} = 1,000 \text{ dm}^3$	$1 \text{ m}^3 = 1,000 \text{ dm}^3$
dm^3 to cm^3	$\text{dm} \times \text{dm} \times \text{dm} = \text{dm}^3$ $1 \text{ dm} = 10 \text{ cm}$ $10 \text{ cm} \times 10 \text{ cm} \times 10 \text{ cm} = 1,000,000 \text{ cm}^3$	$1 \text{ dm}^3 = 1,000 \text{ cm}^3$
cm^3 to mm^3	$\text{cm} \times \text{cm} \times \text{cm} = \text{cm}^3$ $1 \text{ cm} = 10 \text{ mm}$ $10 \text{ mm} \times 10 \text{ mm} \times 10 \text{ mm} = 1,000 \text{ mm}^3$	$1 \text{ cm}^3 = 1,000 \text{ mm}^3$

To have an essential understanding of metric volume, you must be able to identify reasonable values for volume. When testing for reasonableness you should assess both the unit and the unit value. Only by examining both can you determine whether the given volume is reasonable for the situation.

EXAMPLE 9.19

Determining Reasonable Values for Volume

A grandparent wants to send cookies to their grandchild away at college. Which represents a reasonable value for the volume of a box to ship the cookies:

- $3,375 \text{ km}^3$,
- $3,375 \text{ m}^3$, or
- $3,375 \text{ cm}^3$?

 Solution

A volume of $3,375 \text{ km}^3$ is equivalent to a rectangular prism with dimensions of $15 \text{ km} \times 15 \text{ km} \times 15 \text{ km}$, which is far too large for a shipping box. An area of $3,375 \text{ m}^3$ is equivalent to a surface of $15 \text{ m} \times 15 \text{ m} \times 15 \text{ m}$, which is also too large. A reasonable value for the volume of the box is $3,375 \text{ cm}^3$.

> YOUR TURN 9.19

1. Which represents a reasonable value for the volume of a storage area:
8 m³, 8 cm³, or 8 mm³?

EXAMPLE 9.20**Identifying Reasonable Values for Volume**

A food manufacturer is prototyping new packaging for one of its most popular products. Which represents a reasonable value for the volume of the box:

- 2 dm³,
- 2 cm³, or
- 2 mm³?

✓ Solution

A decimeter is equal to 10 centimeters. A box with a volume of 2 dm³ might have the dimensions 1 dm × 1 dm × 2 dm, or 10 cm × 10 cm × 20 cm, which is reasonable. A box with a volume of 2 cm³ or 2 mm³ would be too small.

> YOUR TURN 9.20

1. Which represents a reasonable value for the volume of a fish tank:
40,000 mm³, 40,000 cm³, or 40,000 m³?

EXAMPLE 9.21**Explaining Reasonable Values for Volume**

A farmer has a hay loft. They calculate the volume of the hayloft as 64 cm³. Does the calculation make sense? Explain your answer.

✓ Solution

No. Centimeters are used to determine smaller distances, such as the length of a pencil. A hayloft is more than 64 centimeters long, so a more reasonable unit of value would be m². A volume of 64 m³ can be calculated using the dimensions 4 meters by 4 meters by 4 meters, which are reasonable dimensions for a hayloft. So, a more reasonable value for the volume of the hayloft is 64 m³.

> YOUR TURN 9.21

1. An artist creates a glass paperweight. They decide they want to box and ship the paperweight, so they measure and determine that the volume of the cubic box is 125,000 mm³. Does their calculation make sense? Explain your answer.

Converting Like Units of Measures for Volume

Just like converting units of measure for distance, you can convert units of measure for volume. However, the conversion factor, the number used to multiply or divide to convert from one volume unit to another, is different from the conversion factor for metric distance units. Recall that the conversion factor for volume is exponentially relative to the conversion factor for distance. The most frequently used conversion factors are illustrated in [Figure 9.10](#).

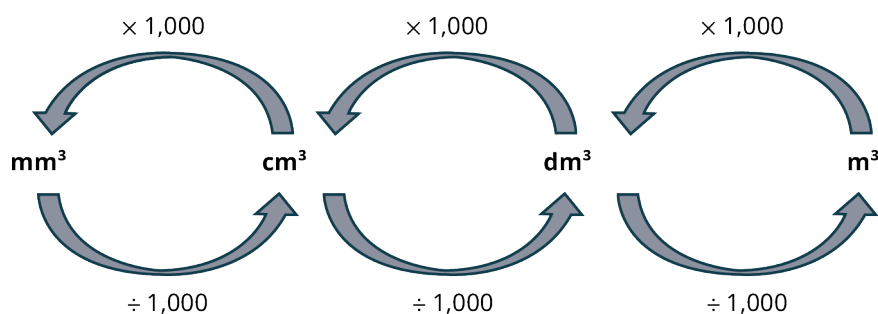


Figure 9.10 Common Conversion Factors for Metric Volume Units

EXAMPLE 9.22

Converting Like Units of Measure for Volume Using Multiplication

A pencil case has a volume of $1,700 \text{ cm}^3$. What is the volume in cubic millimeters?

Solution

Use multiplication to convert from a larger metric volume unit to a smaller metric volume unit. To convert from cm^3 to mm^3 , multiply the value of the volume by 1,000.

$$1,700 \times 1,000 = 1,700,000$$

The pencil case has a volume of $1,700,000 \text{ mm}^3$.

YOUR TURN 9.22

1. A jewelry box has a volume of 8 cm^3 . What is the volume of the jewelry box in cubic millimeters?

VIDEO

[How to Convert Cubic Centimeters to Cubic Meters \(https://openstax.org/r/Convert_CC_to_CM\)](https://openstax.org/r/Convert_CC_to_CM)

EXAMPLE 9.23

Converting Like Units of Measure for Volume Using Multi-Step Multiplication

A shipping container has a volume of 33.2 m^3 . What is the volume in cubic centimeters?

Solution

Use multiplication to convert a larger metric volume unit to a smaller metric volume unit. To convert from m^3 to cm^3 , first multiply the value of the volume by 1,000 to convert from m^3 to dm^3 , and then multiply again by 1,000 to convert from dm^3 to cm^3 .

$$33.2 \times 1,000 \times 1,000 = 33,200,000$$

The shipping container has a volume of $32,200,000 \text{ cm}^3$.

YOUR TURN 9.23

1. A gasoline storage tank has a volume of 37.854 m^3 . What is the volume of the storage tank in cubic centimeters?

EXAMPLE 9.24

Converting Like Units of Measure for Volume Using Multi-Step Division

A holding tank at the local aquarium has a volume of $22,712,000,000 \text{ cm}^3$. What is the volume in cubic meters?

 Solution

Figure 9.10 indicates that when converting from a smaller metric volume unit to a larger metric volume unit you divide using the given conversion factor. To convert from cm^3 to m^3 , divide the value of the volume by 1,000 to first convert from cm^3 to dm^3 , then divide again to convert from dm^3 to m^3 .

$$\text{cm}^3 \text{ to } \text{dm}^3 : \frac{22,712,000,000}{1,000} = 22,712,000$$

$$\text{dm}^3 \text{ to } \text{m}^3 : \frac{22,712,000}{1,000} = 22,712$$

The holding tank has a volume of 22,712 m^3 .

 YOUR TURN 9.24

1. A warehouse has a volume of 465,000,000 cm^3 . What is the volume of the warehouse in cubic meters?

Understanding Other Metric Units of Volume

When was the last time you purchased a bottle of soda? Was the volume of the bottle expressed in cubic centimeters or liters? The liter (L) is a metric unit of capacity often used to express the volume of liquids. A liter is equal in volume to 1 cubic decimeter. A milliliter is equal in volume to 1 cubic centimeter. So, when a doctor orders 10 cc (cubic centimeters) of saline to be administered to a patient, they are referring to 10 mL of saline.

The most frequently used factors for converting from cubic meters to liters are listed in Table 9.2.

m^3 to L	m^3 to mL
1 $\text{dm}^3 = 1 \text{ L}$	1 $\text{dm}^3 = 1,000 \text{ mL}$
1,000 $\text{cm}^3 = 1 \text{ L}$	1 $\text{cm}^3 = 1 \text{ mL}$
1,000,000 $\text{mm}^3 = 1 \text{ L}$	1 $\text{mm}^3 = 0.001 \text{ mL}$

Table 9.2 Relationships Between Metric Volume and Metric Capacity Units

 VIDEO

Converting Metric Units of Volume (https://openstax.org/r/Converting_Metric_Units_of_Volume)

EXAMPLE 9.25

Converting Different Units of Measure for Volume

A holding tank at the local aquarium has a volume of 22,712,000,000 cm^3 . What is the capacity of the holding tank in liters?

 Solution

Use division to convert from cubic centimeters to liters. To determine the equivalent volume in liters, convert from cm^3 to L by dividing the value of the volume in cm^3 by 1,000.

$$\frac{22,712,000,000 \text{ cm}^3}{1,000} = 22,712,000 \text{ L}$$

The holding tank holds 22,712,000 L of water.

> YOUR TURN 9.25

1. A gas can has a volume of $19,000 \text{ cm}^3$. How much gas, in liters, does the gas can hold?

EXAMPLE 9.26**Converting Different Units of Measure for Volume Using Multiplication**

An airplane used 150 m^3 of fuel to fly from New York to Hawaii. How many liters of fuel did the airplane use?

✓ Solution

Because 1 liter is equivalent to 1 cubic decimeter, use multiplication to convert from m^3 to dm^3 . Multiply the value of the volume by 1,000 to convert from m^3 to dm^3 . Because $1 \text{ dm}^3 = 1 \text{ L}$, the resulting value is equivalent to the number of liters used.

$$150 \times 1,000 = 150,000 \text{ dm}^3 = 150,000 \text{ L}$$

The airplane used 150,000 liters of fuel.

> YOUR TURN 9.26

1. A gasoline storage tank has a volume of 37.854 m^3 . What is the volume of the storage tank in liters?

EXAMPLE 9.27**Converting Different Units of Measure for Volume Using Multi-Step Division**

How many liters can a pitcher with a volume of $8,000,000 \text{ mm}^3$ hold?

✓ Solution

Use division to convert from a smaller metric volume unit to a larger metric volume unit. To convert from mm^3 to dm^3 ,

Step 1: Divide by 1,000 to convert from mm^3 to cm^3 .

Step 2: Divide again by 1,000 to convert from cm^3 to dm^3 .

Step 3: Use the unit value to express the volume in terms of liters.

$$\begin{aligned} \frac{8,000,000 \text{ mm}^3}{1,000} &= 8,000 \text{ cm}^3 \\ \frac{8,000 \text{ cm}^3}{1,000} &= 8 \text{ dm}^3 = 8 \text{ L} \end{aligned}$$

The pitcher can hold 8 liters of liquid.

> YOUR TURN 9.27

1. A glass jar has a volume of $800,000 \text{ mm}^3$. How many mL of liquid can the glass jar hold?

Solving Application Problems Involving Volume

Knowing the volume of an object lets you know just how much that object can hold. When making a bowl of punch you might want to know the total amount of liquid a punch bowl can hold. Knowing how many liters of gasoline a car's tank can hold helps determine how many miles a car can drive on a full tank. Regardless of the application, understanding volume is essential to many every day and professional tasks.

EXAMPLE 9.28**Using Volume to Solve Problems**

A cubic shipping carton's dimensions measure $2\text{ m} \times 2\text{ m} \times 2\text{ m}$. A company wants to fill the carton with smaller cubic boxes that measure $10\text{ cm} \times 10\text{ cm} \times 10\text{ cm}$. How many of the smaller boxes will fit in each shipping carton?

 Solution

Step 1: Determine the volume of the shipping carton.

$$2\text{ m} \times 2\text{ m} \times 2\text{ m} = 8\text{ m}^3$$

Step 2: Use the appropriate conversion factor to convert the volume of the shipping carton from m^3 to cm^3 .

$$\begin{aligned} 8\text{ m}^3 \times 1,000 &= 8,000\text{ dm}^3 \\ 8,000\text{ dm}^3 \times 1,000 &= 8,000,000\text{ cm}^3 \end{aligned}$$

Step 3: Determine the volume of the smaller boxes.

$$10\text{ cm} \times 10\text{ cm} \times 10\text{ cm} = 1,000\text{ cm}^3$$

Step 4: Divide the volume of the shipping carton, in cm^3 , by the volume of the smaller box, in cm^3 .

$$\frac{8,000,000\text{ cm}^3}{1,000\text{ cm}^3} = 8,000$$

The shipping carton will hold 8,000 smaller boxes.

**YOUR TURN 9.28**

1. A factory can mill 300 cubic meters of flour each day. They package the flour in boxes that measure $20\text{ cm} \times 5\text{ cm} \times 30\text{ cm}$. How many boxes of flour does the factory produce each day?

EXAMPLE 9.29**Solving Volume Problems with Different Units**

A carton of juice measures 6 cm long, 6 cm wide and 20 cm tall. A factory produces 28,800 liters of orange juice each day. How many cartons of orange juice are produced each day?

 Solution

Step 1: Find the volume of the carton in cubic centimeters.

$$6\text{ cm} \times 6\text{ cm} \times 20\text{ cm} = 720\text{ cm}^3$$

Step 2: Convert the volume in cm^3 to liters.

$$\begin{aligned} 1\text{ cm}^3 &= 1\text{ mL} \\ 720\text{ cm}^3 &= 720\text{ mL} \\ \frac{720\text{ mL}}{1,000} &= 0.72\text{ L} \end{aligned}$$

Step 3: Divide the number of liters of orange juice produced each day by the volume of each carton.

$$\frac{28,800 \text{ L}}{0.72 \text{ L}} = 40,000$$

The factory produces 40,000 cartons of orange juice each day.

> YOUR TURN 9.29

1. An ice cream maker boxes frozen yogurt mix in boxes that measure 25 cm long, 8 cm wide and 35 cm tall. They produce 42,000 liters of frozen yogurt mix each day. How many boxes of frozen yogurt mix are produced each day?

EXAMPLE 9.30

Solving Complex Volume Problems

A fish tank measures 60 cm long, 15 cm wide and 34 cm tall (Figure 9.11). The tank is 25 percent full. How many liters of water are needed to completely fill the tank?

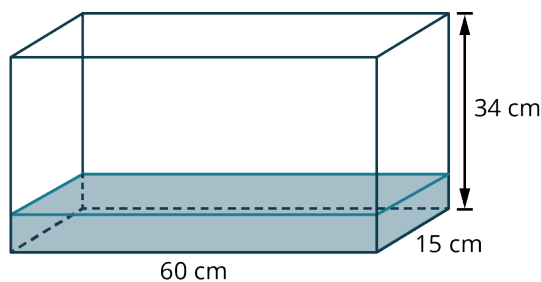


Figure 9.11

✓ Solution

Step 1: Determine the volume of the fish tank in cubic centimeters.

$$60 \text{ cm} \times 15 \text{ cm} \times 34 \text{ cm} = 30,600 \text{ cm}^3$$

Step 2: Convert the volume in cm^3 to volume in liters.

$$\begin{aligned} 1 \text{ cm}^3 &= 1 \text{ mL} \\ 30,600 \text{ cm}^3 &= 30,600 \text{ mL} \\ \frac{30,600 \text{ mL}}{1,000} &= 30.6 \text{ L} \end{aligned}$$

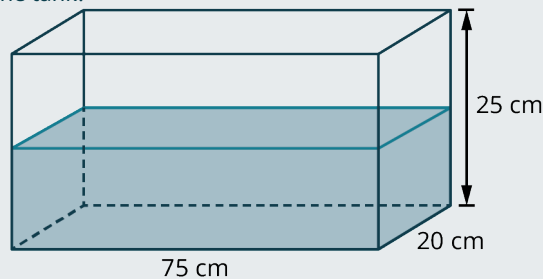
Step 3: Since the tank is 25 percent full, the tank is 75 percent empty. Convert 75 percent to its decimal equivalent. Multiply the total volume by 75 percent expressed in decimal form to determine how many liters of water are required to fill the tank.

$$\begin{aligned} 75\% &= 0.75 \\ 30.6 \times 0.75 &= 22.95 \end{aligned}$$

So, 22.95 liters of water are needed to fill the tank.

YOUR TURN 9.30

1. A fish tank measures 75 cm long, 20 cm wide and 25 cm tall. The tank is 50 percent full. How many liters of water are needed to completely fill the tank?



WORK IT OUT

How Does Shape Affect Volume?

Take two large sheets of card stock. Roll one piece to tape the longer edges together to make a cylinder. Tape the cylinder to the other piece of card stock which serves as the base of the cylinder. Fill the cylinder to the top with cereal. Pour the cereal from the cylinder into a plastic storage or shopping bag. Remove the cylinder from the base and the tape from the cylinder. Re-roll the cylinder along the shorter edges a tape together. Attach the new cylinder to the base. Pour the cereal from the plastic bag into the cylinder. What do you observe? How does the shape of a container affect its volume?

Check Your Understanding

For the following exercises, determine the most reasonable value for each volume.

16. Terrarium: $50,000 \text{ km}^3$, $50,000 \text{ m}^3$, $50,000 \text{ cm}^3$, or $50,000 \text{ mm}^3$
17. Milk carton: 236,000 L, 236 L, 236,000 mL, or 236 mL
18. Box of crackers: $1,500 \text{ km}^3$, $1,500 \text{ m}^3$, $1,500 \text{ cm}^3$, or $1,500 \text{ mm}^3$

For the following exercises, Convert the given volume to the units shown.

19. $42,500 \text{ mm}^3 = \underline{\hspace{2cm}} \text{ cm}^3$
20. $1.5 \text{ dm}^3 = \underline{\hspace{2cm}} \text{ mL}$
21. $6.75 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

For the following exercises, determine the volume of objects with the dimensions shown.

22. $20 \text{ cm} \times 20 \text{ m} \times 20 \text{ cm}$
 $V = \underline{\hspace{2cm}} \text{ L}$
23. $12 \text{ mm} \times 5 \text{ cm} \times 1.7 \text{ cm}$
 $V = \underline{\hspace{2cm}} \text{ mL}$
24. $7.3 \text{ m} \times 3.2 \text{ m} \times 7 \text{ m}$
 $V = \underline{\hspace{2cm}} \text{ m}^3$



SECTION 9.3 EXERCISES

For the following exercises, determine the most reasonable value for each volume.

1. Fish tank:
 $71,120 \text{ km}^3$, $71,120 \text{ m}^3$, $71,120 \text{ cm}^3$, or $71,120 \text{ mm}^3$
2. Juice box:
125,000 L, 125 L, 125,000 mL, or 125 mL
3. Box of cereal:
 $2,700 \text{ km}^3$, $2,700 \text{ m}^3$, $2,700 \text{ cm}^3$, or $2,700 \text{ mm}^3$
4. Water bottle:
5 L, 0.5 L, 5 mL, or 0.5 mL
5. Shoe box:

3,600 km³, 3.6 m³, 3,600 cm³, or 3,600 mm³

6. Swimming pool:

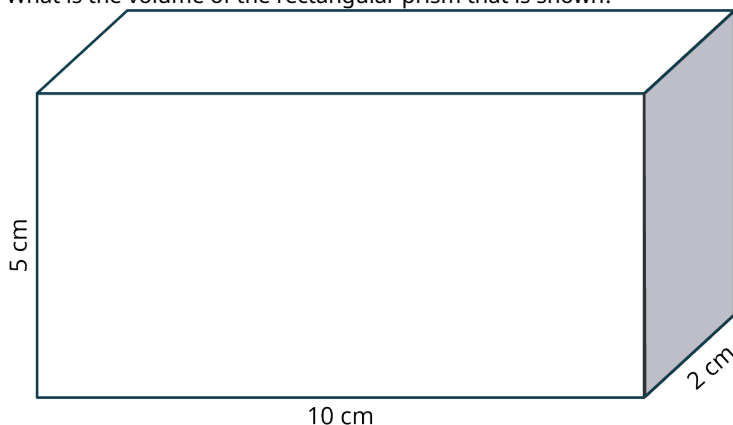
45 L, 45,000 L, 45 mL, or 45,000 mL

For the following exercises, convert the given volume to the units shown.

7. 38,861 mm³ = _____ cm³
8. 13 dm³ = _____ mL
9. 874 cm³ = _____ mm³
10. 4 m³ = _____ cm³
11. 0.00003 m³ = _____ mm³
12. 57,500 mm³ = _____ L
13. 0.007 m³ = _____ L
14. 8,600 cm³ = _____ m³
15. 45.65 m³ = _____ cm³
16. 0.06 m³ = _____ dm³
17. 0.081 m³ = _____ mL
18. 3,884,000 mm³ = _____ m³

For the following exercises, determine the volume of objects with the dimensions shown.

19. 30 cm × 20 m × 10 cm
 $V =$ _____ L
20. 17 mm × 3 cm × 2.5 cm
 $V =$ _____ mL
21. 3.4 m × 2.5 m × 10 m
 $V =$ _____ m³
22. 325 mm × 20 cm × 0.05 m
 $V =$ _____ cm³
23. 3.7 m × 4 m × 5.5 m
 $V =$ _____ m³
24. 18 dm × 0.8 m × 150 cm
 $V =$ _____ L
25. 15 cm × 400 mm × 3 dm
 $V =$ _____ mL
26. 3.5 cm × 200 mm × 0.7 dm
 $V =$ _____ cm³
27. 35 m × 1.2 m × 0.007 km
 $V =$ _____ m³
28. A box has dimensions of 20 cm × 15 cm × 30 cm. The box currently holds 1,250 cm³ of rice. How many cubic centimeters of rice are needed to completely fill the box?
29. The dimensions of a medium storage unit are 4 m × 4 m × 8 m. What is the volume of a small storage area with dimensions half the size of the medium unit?
30. How much liquid, in liters, can a container with dimensions of 40 cm × 20 cm × 120 cm hold?
31. What is the volume of the rectangular prism that is shown?



32. A box is 15 centimeters long and 5 centimeters wide. The volume of the box is 225 cm^3 . What is the height of the box?
33. Kareem mixed two cartons of orange juice, three 2-liter bottles of soda water and six cans of cocktail fruits to make a fruit punch for a party. The cartons of orange juice and cans of cocktail fruits each have a volume of 500 cm^3 . How much punch, in liters, did Kareem make?
34. A holding tank has dimensions of $16 \text{ m} \times 8 \text{ m} \times 8 \text{ m}$. If the tank is half-full, how more liters of liquid can the tank hold?
35. A large plastic storage bin has dimensions of $16 \text{ cm} \times 16 \text{ cm} \times 16 \text{ cm}$. A medium bin's dimensions are half the size of the large bin. A small bin's dimensions are the size of the medium bin. If the storage bins come in a set of 3—small, medium, and large—what is the total volume of the storage bin set in cubic centimeters?
36. A soft serve ice cream machine holds a 19.2 liter bag of ice cream mix. If the average serving size of an ice cream cone is 120 mL, how many cones can be made from each bag of mix?
37. A shipping carton has dimensions of $0.5 \text{ m} \times 0.5 \text{ m} \times 0.5 \text{ m}$. How many boxes with dimensions of $50 \text{ mm} \times 50 \text{ mm} \times 50 \text{ mm}$ will fit in the shipping carton?
38. A recipe for chili makes 3.5 liters of chili. If a restaurant serves chili in 250 mL bowls, how many bowls of chili can they serve?
39. A contractor is building an in-ground pool. They excavate a pit that measures $12 \text{ m} \times 9 \text{ m} \times 2.5 \text{ m}$. The dirt is being taken away in a truck that holds 30 m^3 . How many trips will the truck have to make to cart away all of the dirt?
40. A juice dispenser measures $30 \text{ cm} \times 30 \text{ cm} \times 30 \text{ cm}$. How many 375 mL servings will a full dispenser serve?

9.4 Measuring Weight



Figure 9.12 Weight scale at the local Antigua market (credit: “Weight scale at the local Antigua market” by Lucía García González/Flickr, CC0 1.0 Public Domain Dedication)

Learning Objectives

After completing this section, you should be able to:

1. Identify reasonable values for weight applications.
2. Convert units of measures of weight.
3. Solve application problems involving weight.

In the metric system, weight is expressed in terms of grams or kilograms, with a kilogram being equal to 1,000 grams. A paper clip weighs about 1 gram. A liter of water weighs about 1 kilogram. In fact, in the same way that 1 liter is equal in volume to 1 cubic decimeter, the kilogram was originally defined as the mass of 1 liter of water. In some cases, particularly in scientific or medical settings where small amounts of materials are used, the milligram is used to express weight. At the other end of the scale is the metric ton (mt), which is equivalent to 1,000 kilograms. The average car weighs about 2 metric tons.

Any discussion about metric weight must also include a conversation about **mass**. Scientifically, mass is the amount of

matter in an object whereas weight is the force exerted on an object by gravity. The amount of mass of an object remains constant no matter where the object is. Identical objects located on Earth and on the moon will have the same mass, but the weight of the objects will differ because the moon has a weaker gravitational force than Earth. So, objects with the same mass will weigh less on the moon than on Earth.

Since there is no easy way to measure mass, and since gravity is just about the same no matter where on Earth you go, people in countries that use the metric system often use the words mass and weight interchangeably. While scientifically the kilogram is only a unit of mass, in everyday life it is often used as a unit of weight as well.

Reasonable Values for Weight

To have an essential understanding of metric weight, you must be able to identify reasonable values for weight. When testing for reasonableness, you should assess both the unit and the unit value. Only by examining both can you determine whether the given weight is reasonable for the situation.

VIDEO

[Metric System: Units of Weight \(https://openstax.org/r/Metric_System:_units_of_weight\)](https://openstax.org/r/Metric_System:_units_of_weight)

EXAMPLE 9.31

Identifying Reasonable Units for Weight

Which is the more reasonable value for the weight of a newborn baby:

- 3.5 kg or
- 3.5 g?

Solution

Using our reference weights, a baby weighs more than 3.5 paperclips, so 3.5 kilograms is a more reasonable value for the weight of a newborn baby.

YOUR TURN 9.31

1. Which represents a reasonable value for the weight of a penny:
2.5 g or 2.5 kg

EXAMPLE 9.32

Determining Reasonable Values for Weight

Which of the following represents a reasonable value for the weight of three lemons?

- 250 g,
- 2,500 g, or
- 250 kg?

Solution

Because a kilogram is about 2.2 pounds, we can eliminate 250 kg as it is way too heavy. 2,500 grams is equivalent to 2.5 kilograms, or about five pounds, which is again, too heavy. So, a reasonable value for the weight of three lemons would be 250 grams.

YOUR TURN 9.32

1. Which represents a reasonable value for the weight of a car:
1,300 g, 130 kg, or 1,300 kg?

?? WHO KNEW?**How Do You Measure the Weight of a Whale?**

It is impossible to weigh a living whale. Fredrik Christiansen from the Aarhus Institute of Advanced Studies in Denmark developed an innovative way to measure the weight of whales. Using images taken from a drone and computer modeling, the weight of a whale can be estimated with great accuracy.

▶ VIDEO

[Using Drones to Weigh Whales? \(https://openstax.org/r/Using_drones_to_weigh_whales?\)](https://openstax.org/r/Using_drones_to_weigh_whales?)

EXAMPLE 9.33**Explaining Reasonable Values for Weight**

The blue whale is the largest living mammal on Earth. Which of the following is a reasonable value for the weight of a blue whale: 149 g, 149 kg, or 149 mt? Explain your answer.

✓ Solution

A reasonable value for the weight of a blue whale is 149 metric tons. Both 149 g and 149 kg are much too small a value for the largest living mammal on Earth.

> YOUR TURN 9.33

1. The Etruscan shrew is one of the world's smallest mammals. It has a huge appetite, eating almost twice its weight in food each day. Its heart beats at a rate of 25 beats per second! Which of the following is a reasonable value for the weight of an Etruscan shrew: 2 g, 2 kg, or 2 mt? Explain your answer.

Converting Like Units of Measures for Weight

Just like converting units of measure for distance, you can convert units of measure for weight. The most frequently used conversion factors for metric weight are illustrated in [Figure 9.13](#).

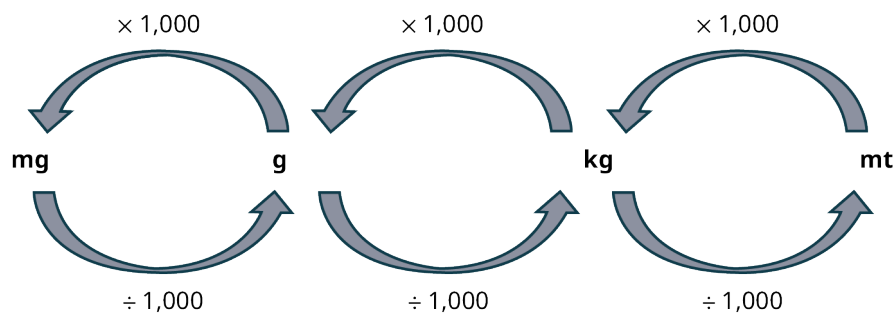


Figure 9.13 Common Conversion Factors for Metric Weight Units

EXAMPLE 9.34**Converting Metric Units of Weight Using Multistep Division**

How many kilograms are in 24,300,000 milligrams?

✓ Solution

Use division to convert from a smaller metric weight unit to a larger metric weight unit. To convert from milligrams to kilograms,

Step 1: Divide the value of the weight in milligrams by 1,000 to first convert from milligrams to grams.

Step 2: Divide by 1,000 again to convert from grams to kilograms.

$$\frac{24,300,000 \text{ mg}}{1,000} = 24,300 \text{ g}$$

$$\frac{24,300 \text{ g}}{1,000} = 24.3 \text{ kg}$$

So, 24,300,000 milligrams are equivalent to 24.3 kilograms.

YOUR TURN 9.34

1. How many kilograms are in 175,000 milligrams?

EXAMPLE 9.35

Converting Metric Units of Weight Using Multiplication

The average ostrich weighs approximately 127 kilograms. How many grams does an ostrich weigh?

Solution

Use multiplication to convert from a larger metric weight unit to a smaller metric weight unit. To convert from kilograms to grams, multiply the value of the weight by 1,000.

$$127 \text{ kg} \times 1,000 = 127,000 \text{ g}$$

The average ostrich weighs 127,000 grams.

YOUR TURN 9.35

1. The world's heaviest tomato weighed 4.869 kg when measured on July 15, 2020. How much did the tomato weigh in grams?

EXAMPLE 9.36

Converting Metric Units of Weight Using Multistep Multiplication

How many milligrams are there in 0.025 kilograms?

Solution

Use multiplication to convert from a larger metric weight unit to a smaller metric weight unit. To convert from kilograms to grams,

Step 1: Multiply the value of the weight by 1,000.

Step 2: Multiply the result by 1,000 to convert from grams to milligrams.

$$0.025 \text{ kg} \times 1,000 = 25 \text{ g}$$

$$25 \text{ g} \times 1,000 = 25,000 \text{ mg}$$

So, 0.025 kilograms is equivalent to 25,000 milligrams.

> YOUR TURN 9.36

1. How many milligrams are there in 1.23 kilograms?

▶ VIDEO

[Metric Units of Mass: Convert mg, g, and kg \(https://openstax.org/r/Convert_mg-g-kg\)](https://openstax.org/r/Convert_mg-g-kg)

Solving Application Problems Involving Weight

From children's safety to properly cooking a pie, knowing how to solve problems involving weight is vital to everyday life. Let's review some ways that knowing how to work with metric weight can facilitate important decisions and delicious eating.

EXAMPLE 9.37

Comparing Weights to Solve Problems

The maximum weight for a child to safely use a car seat is 29 kilograms. If a child weighs 23,700 grams, can the child safely use the car seat?

✓ Solution

Step 1: Convert the child's weight in grams to kilograms.

$$23,700 \text{ g} \div 1,000 = 23.7 \text{ kg}$$

Step 2: Compare the two weights.

$$23.7 \text{ kg} < 29 \text{ kg}$$

Yes, the child can safely use the car seat.

> YOUR TURN 9.37

1. The dosage recommendations for a popular brand of acetaminophen are listed in table below. What is the recommended dosage for a child who weighs 17,683 grams?

Weight	Dosage
11 kg to 15 kg	5 mL
16 kg to 21 kg	7.5 mL
22 kg to 27 kg	10 mL

EXAMPLE 9.38

Solving Multistep Weight Problems

A recipe for scones calls for 350 grams of flour. How many kilograms of flour are required to make 4 batches of scones?

✓ Solution

Step 1: Multiply the grams of flour need by 4 to determine the total amount of flour needed.

$$350 \text{ g} \times 4 = 1,400 \text{ g}$$

Step 2: Convert from grams to kilograms.

$$\frac{1,400 \text{ g}}{1,000} = 1.4 \text{ kg}$$

So, 1.4 kilograms of flour are needed to make four batches of scones.

YOUR TURN 9.38

1. A croissant recipe calls for 500 g of flour. How many kilograms of flour are required to make 10 batches of croissants?

EXAMPLE 9.39

Solving Complex Weight Problems

The average tomato weighs 140 grams. A farmer needs to order boxes to pack and ship their tomatoes to local grocery stores. They estimate that this year's harvest will yield 125,000 tomatoes. A box can hold 12 kilograms of tomatoes. How many boxes does the farmer need?

Solution

Step 1: Determine the total estimated weight of the harvested tomatoes.

$$140 \text{ g} \times 125,000 = 17,500,000 \text{ g}$$

Step 2: Convert the total weight from grams to kilograms.

$$17,500,000 \text{ g} \div 1,000 = 17,500 \text{ kg}$$

Step 3: Divide the weight of the tomatoes by the weight each box can hold.

$$17,500 \text{ kg} \div 12 \text{ kg} \approx 1,458$$

So, the farmer will need to order 1,458 boxes.

YOUR TURN 9.39

1. The average potato weighs 225 grams. A grocery chain orders 5,000 bags of potatoes. Each bag weighs 5 kg. Approximately how many potatoes did they order?

Check Your Understanding

For the following exercises, determine the most reasonable value for each weight.

25. Candy bar:
50 kg, 50 g, or 50 mg
26. Lion:
180 kg, 180 g, or 180 mg
27. Basketball:
624 kg, 624 g, or 624 mg

For the following exercises, convert the given weight to the units shown.

28. 8,900 g = _____ kg

29. $17 \text{ g} = \underline{\hspace{2cm}} \text{ mg}$

30. $0.07 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$

For the following exercises, determine the total weight in the units shown.

31. three 48 g granola bars $\underline{\hspace{2cm}} \text{ kg}$

32. seven 28 g cheese slices $\underline{\hspace{2cm}} \text{ mg}$

33. six 15 mg tea bags $\underline{\hspace{2cm}} \text{ g}$



SECTION 9.4 EXERCISES

For the following exercises, determine the most reasonable value for each weight.

1. Aspirin tablet:
300 kg, 300 g, or 300 mg
2. Elephant:
5,000 kg, 5,000 g, or 5,000 mg
3. Baseball:
145 kg, 145 g, or 145 mg
4. Orange:
115 kg, 115 g, or 115 mg
5. Pencil:
6 kg, 6 g, or 6 mg
6. Automobile:
1,300 kg, 1,300 g, or 1,300 mg

For the following exercises, convert the given weight to the units shown.

7. $3,500 \text{ g} = \underline{\hspace{2cm}} \text{ kg}$
8. $53 \text{ g} = \underline{\hspace{2cm}} \text{ mg}$
9. $0.02 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$
10. $200 \text{ mg} = \underline{\hspace{2cm}} \text{ g}$
11. $2.3 \text{ g} = \underline{\hspace{2cm}} \text{ mg}$
12. $20 \text{ kg} = \underline{\hspace{2cm}} \text{ mg}$
13. $2,300 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$
14. $8,700 \text{ mg} = \underline{\hspace{2cm}} \text{ g}$
15. $9,730 \text{ mg} = \underline{\hspace{2cm}} \text{ kg}$
16. $0.0078 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$
17. $2.34 \text{ g} = \underline{\hspace{2cm}} \text{ mg}$
18. $234.5 \text{ mg} = \underline{\hspace{2cm}} \text{ g}$

For the following exercises, determine the total weight in the units shown.

19. Three 350 mg tablets $\underline{\hspace{2cm}} \text{ g}$
20. Seven 115 g soap bars $\underline{\hspace{2cm}} \text{ kg}$
21. Six 24 g batteries $\underline{\hspace{2cm}} \text{ mg}$
22. Fifty 3.56 g pennies $\underline{\hspace{2cm}} \text{ mg}$
23. Eight 2.25 kg bags of potatoes $\underline{\hspace{2cm}} \text{ g}$
24. Four 23 kg sacks of flour $\underline{\hspace{2cm}} \text{ g}$
25. Ten 2.5 kg laptops $\underline{\hspace{2cm}} \text{ g}$
26. Seven 1,150 g chickens $\underline{\hspace{2cm}} \text{ kg}$
27. Ninety 4,500 mg marbles $\underline{\hspace{2cm}} \text{ kg}$
28. There are 26 bags of flour. Each bag weighs 5 kg. What is the total weight of the flour?
29. The average female hippopotamus weighs 1,496 kg. The average male hippopotamus weighs 1,814 kg. How much heavier, in grams, is the male hippopotamus than the female hippopotamus?
30. Twelve pieces of cardboard weigh 72 grams. What is the weight of one piece of cardboard?
31. Miguel's backpack weighs 2.4 kg and Shanayl's backpack weighs 2,535 grams. Whose backpack is heavier and by how much?
32. A souvenir chocolate bar weighs 1.815 kg. If you share the candy bar equally with two friends, how many grams of chocolate does each person get?

33. You purchase 10 bananas that weigh 50 grams each. If bananas cost \$5.50 per kilogram, how much did you pay?
34. A family-size package of ground meat costs \$15.75. The package weighs 4.5 kg. What is the cost per gram of the meat?
35. A box containing 6 identical books weighs 7.2 kg. The box weighs 600 g. What is the weight of each book in grams?
36. A 2.316 kg bag of candy is equally divided into 12 party bags. What is the weight of the candy, in grams, in each party bag?
37. A store has 450 kg of flour at the beginning of the day. At the end of the day the store has 341 kg of flour. If flour costs \$0.35 per kilogram, how much flour, in dollars, did the store sell that day?
38. A local restaurant offers lobster for \$110 per kilogram. What is the price for a lobster that weighs 450 grams?
39. A student's backpack weighs 575 grams. Their books weigh 3.5 kg. If the student's weight while wearing their backpack is 58.25 kg, how much does the student weigh in kilograms?
40. The weight of a lamb is 41 kg 340 g. What is the total weight, in kilograms, of four lambs of the same weight?

9.5 Measuring Temperature

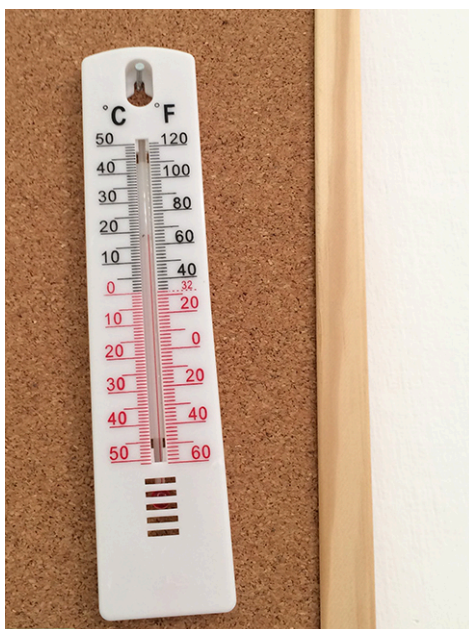


Figure 9.14 A thermometer that measures temperature in both customary and metric units. (credit: “Thermometer” by Jeff Djevdet/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Convert between Fahrenheit and Celsius.
2. Identify reasonable values for temperature applications.
3. Solve application problems involving temperature.

When you touch something and it feels warm or cold, what is that really telling you about that substance? **Temperature** is a measure of how fast atoms and molecules are moving in a substance, whether that be the air, a stove top, or an ice cube. The faster those atoms and molecules move, the higher the temperature.

In the metric system, temperature is measured using the Celsius (°C) scale. Because temperature is a condition of the physical properties of a substance, the Celsius scale was created with 100 degrees separating the point at which water freezes, 0 °C, and the point at which water boils, 100 °C. Scientifically, these are the points at which water molecules change from one state of matter to another—from solid (ice) to liquid (water) to gas (water vapor).

 When reading temperatures, it's important to look beyond the degree symbol to determine which temperature scale the units express. For example, 13 °C reads “13 degrees Celsius,” indicating that the temperature is expressed using the Celsius scale, while 13 °F reads “13 degrees Fahrenheit,” indicating that the temperature is expressed using the

Fahrenheit scale.

 VIDEO

[Misconceptions About Temperature \(https://openstax.org/r/Misconceptions_About_Temperature\)](https://openstax.org/r/Misconceptions_About_Temperature)

 WHO KNEW?

How Many Temperature Scales Are There?

Did you know that in addition to Fahrenheit and Celsius, there is a third temperature scale widely used throughout the world? The Kelvin scale starts at absolute zero, the lowest possible temperature at which there is no heat energy present at all. It is primarily used by scientists to measure very high or very low temperatures when water is not involved.

Converting Between Fahrenheit and Celsius Temperatures

Understanding how to convert between Fahrenheit and Celsius temperatures is an essential skill in understanding metric temperatures. You likely know that below 32 °F means freezing temperatures and perhaps that the same holds true for 0 °C. While it may be difficult to recall that water boils at 212 °F, knowing that it boils at 100 °C is a fairly easy thing to remember.

But what about all the temperatures in between? What is the temperature in degrees Celsius on a scorching summer day? What about a cool autumn afternoon? If a recipe instructs you to preheat the oven to 350 °F, what Celsius temperature do you set the oven at?

[Figure 9.15](#) lists common temperatures on both scales, because we don't use Celsius temperatures daily it's difficult to remember them. Fortunately, we don't have to. Instead, we can convert temperatures from Fahrenheit to Celsius and from Celsius to Fahrenheit using a simple algebraic expression.

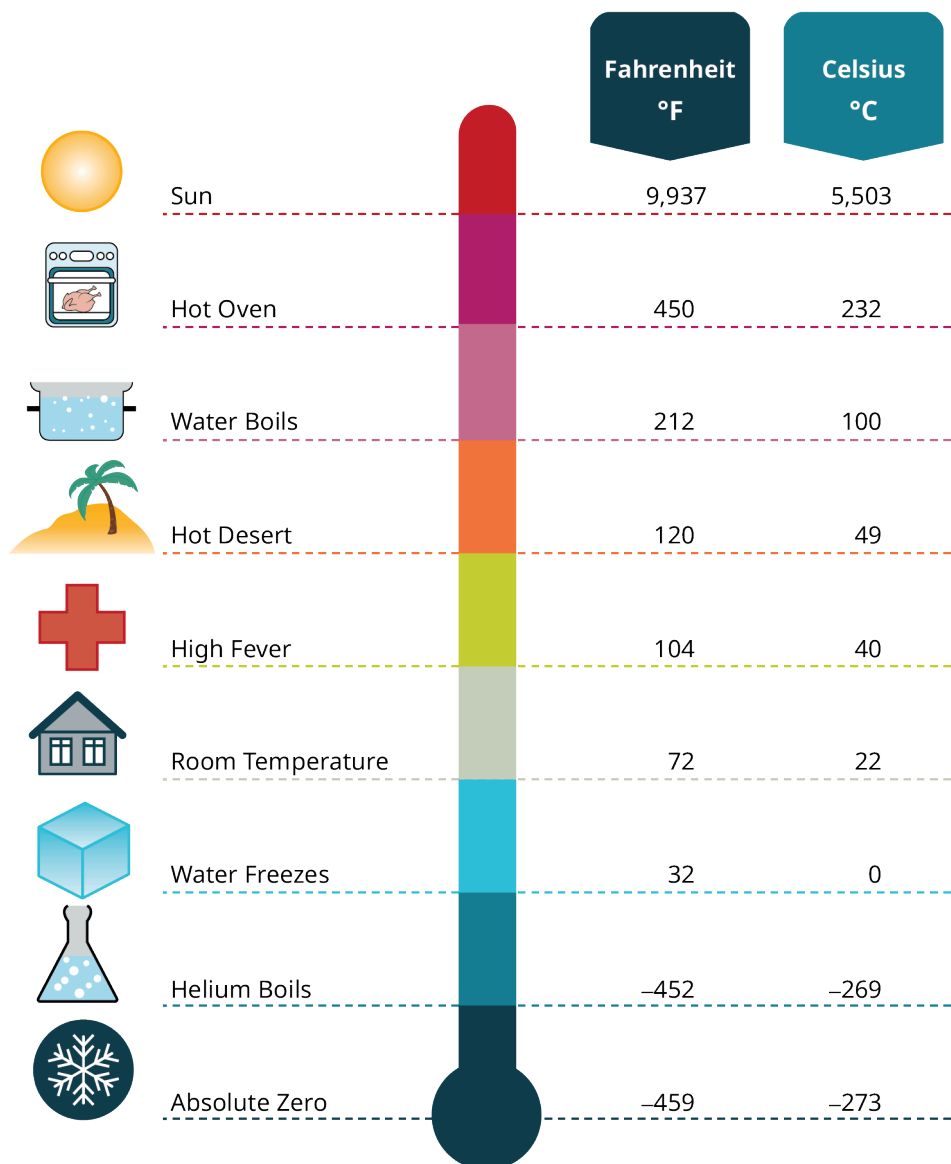


Figure 9.15 Common Temperatures

FORMULA

The formulas used to convert temperatures from Fahrenheit to Celsius or from Celsius to Fahrenheit are outlined in

Table 9.3.

Fahrenheit to Celsius	Celsius to Fahrenheit
$C = \frac{5}{9}(F - 32)$	$F = \frac{9}{5}C + 32$

Table 9.3 Temperature Conversion Formulas

EXAMPLE 9.40**Converting Temperatures from Fahrenheit to Celsius**

A recipe calls for the oven to be set to 392 °F. What is the temperature in Celsius?

✓ Solution

Use the formula in [Table 9.3](#) to convert from Fahrenheit to Celsius.

$$\begin{aligned}
 C &= \frac{5}{9}(F - 32) \\
 C &= \frac{5}{9}(392 - 32) \\
 C &= \frac{5}{9}(360) \\
 C &= 200
 \end{aligned}$$

So, 392 °F is equivalent to 200 °C.

> YOUR TURN 9.40

1. What is 482 °F in Celsius?

EXAMPLE 9.41**Converting Temperatures from Celsius to Fahrenheit**

On a sunny afternoon in May, the temperature in London was 20 °C. What was the temperature in degrees Fahrenheit?

✓ Solution

Use the formula in [Table 9.3](#) to convert from Celsius to Fahrenheit.

$$\begin{aligned}
 F &= \frac{9}{5}C + 32 \\
 F &= \frac{9}{5}(20) + 32 \\
 F &= 36 + 32 \\
 F &= 68
 \end{aligned}$$

The temperature was 68 °F.

> YOUR TURN 9.41

1. What is 75 °C in Fahrenheit?

EXAMPLE 9.42**Comparing Temperatures in Celsius and Fahrenheit**

A manufacturer requires a vaccine to be stored in a refrigerator at temperatures between 36 °F and 46 °F. The refrigerator in the local pharmacy cools to 3 °C. Can the vaccine be stored safely in the pharmacy's refrigerator?

✓ Solution

Use the formula in [Table 9.3](#) to convert from Celsius to Fahrenheit.

$$\begin{aligned} F &= \frac{9}{5}C + 32 \\ F &= \frac{9}{5}(3) + 32 \\ F &= 5.4 + 32 \\ F &= 37.4 \end{aligned}$$

Then, compare the temperatures.

$$36\text{ °F} < 37.4\text{ °F} < 46\text{ °F}$$

Yes. 37.4 °F falls within the acceptable range to store the vaccine, so it can be stored safely in the pharmacy's refrigerator.

> YOUR TURN 9.42

1. In July, the average temperature in Madrid is 16.7 °C. The average temperature in Toronto in July is 57.4 °F. Which city has the higher temperature, and by how many degrees Celsius?

Reasonable Values for Temperature

While knowing the exact temperature is important in most cases, sometimes an approximation will do. When trying to assess the reasonableness of values for temperature, there is a quicker way to convert temperatures for an approximation using mental math. These simpler formulas are listed in [Table 9.4](#).

FORMULA

The formulas used to estimate temperatures from Fahrenheit to Celsius or from Celsius to Fahrenheit are outlined in [Table 9.4](#).

Fahrenheit to Celsius	Celsius to Fahrenheit
$C = \frac{F-30}{2}$	$F = 2C + 30$

Table 9.4 Estimate Temperature Conversion

▶ VIDEO

[Temperature Conversion Trick \(https://openstax.org/r/Temperature_Conversion_Trick\)](https://openstax.org/r/Temperature_Conversion_Trick)

EXAMPLE 9.43**Using Benchmark Temperatures to Determine Reasonable Values for Temperatures**

Which is the more reasonable value for the temperature of a freezer?

- 5 °C or
- -5 °C?

 Solution

We know that water freezes at 0 °C. So, the more reasonable value for the temperature of a freezer is −5 °C, which is below 0 °C. At temperature of 5 °C is above freezing.

 YOUR TURN 9.43

1. To make candy apples, you must boil the sugar mixture for about 20 minutes. Which is the more reasonable value for the temperature of the mixture when boiling:
 - 48 °C or
 - 148 °C?

EXAMPLE 9.44**Using Estimation to Determine Reasonable Values for Temperatures**

The average body temperature is generally accepted as 98.6 °F. What is a reasonable value for the average body temperature in degrees Celsius:

- 98.6 °C,
- 64.3 °C, or
- 34.3 °C?

 Solution

To estimate the average body temperature in degrees Celsius, subtract 30 from the temperature in degrees Fahrenheit, and divide the result by 2.

$$\frac{(98.6 - 30)}{2} = \frac{68.6}{2} = 34.3$$

A reasonable value for average body temperature is 34.3 °C.

 YOUR TURN 9.44

1. Which represents a reasonable value for temperature of a hot summer day:
 - 102 °C,
 - 37 °C, or
 - 20 °C?

EXAMPLE 9.45**Using Conversion to Determine Reasonable Values for Temperatures**

Which is a reasonable temperature for storing chocolate:

- 28 °C,
- 18 °C, or
- 2 °C?

 Solution

Use the formula in [Table 9.3](#) to determine the temperature in degrees Fahrenheit.

$$F = \frac{9}{5}C + 32$$

$$F = \frac{9}{5}(28) + 32 = 82.4$$

$$F = \frac{9}{5}(18) + 32 = 64.4$$

$$F = \frac{9}{5}(2) + 32 = 35.6$$

A temperature of 82.4 °F would be too hot, causing the chocolate to melt. A temperature of 35.6 °F is very close to freezing, which would affect the look and feel of the chocolate. So, a reasonable temperature for storing chocolate is 18 °C, or 64.4 °F.

> YOUR TURN 9.45

- Which represents a reasonable temperature for cooking chili:
 - 240 °C,
 - 60 °C, or
 - 6 °C?

Solving Application Problems Involving Temperature

Whether traveling abroad or working in a clinical laboratory, knowing how to solve problems involving temperature is an important skill to have. Many food labels express sizes in both ounces and grams. Most rulers and tape measures are two-sided with one side marked in inches and feet and the other in centimeters and meters. And while many thermometers have both Fahrenheit and Celsius scales, it really isn't practical to pull out a thermometer when cooking a recipe that uses metric units. Let's review a few instances where knowing how to fluently use the Celsius scale helps solve problems.

EXAMPLE 9.46

Using Subtraction to Solve Temperature Problems

The temperature in the refrigerator is 4 °C. The temperature in the freezer is 21 °C lower. What is the temperature in the freezer?

✓ Solution

Use subtraction to find the difference.

$$4 - 21 = -17$$

So, the temperature in the freezer is -17 °C.

> YOUR TURN 9.46

- At 6 PM, the temperature was 4 °C. By 6 AM the temperature had fallen by 6 °C. What was the temperature at 6 AM?

EXAMPLE 9.47

Using Addition to Solve Temperature Problems

A scientist was using a liquid that was 35 °C. They needed to heat the liquid to raise the temperature by 6 °C. What was the temperature after the scientist heated it?

✓ Solution

Use addition to find the new temperature.

$$35 + 6 = 41$$

The temperature of the liquid was 41 °C after the scientist heated it.

YOUR TURN 9.47

1. A hot dog was cooked in a microwave oven. The hot dog was 4 °C when it was put in the microwave and the temperature increased by 59 °C when it was taken out. What was the temperature of the cooked hot dog?

EXAMPLE 9.48

Solving Complex Temperature Problems

The optimum temperature for a chemical compound to develop its unique properties is 392 °F. When the heating process begins, the temperature of the compound is 20 °C. For safety purposes the compound can only be heated 9 °C every 15 minutes. How long until the compound reaches its optimum temperature?

Solution

Step 1: Determine the optimum temperature in degrees Celsius using the formula in [Table 9.3](#).

$$\begin{aligned}C &= \frac{5}{9}(F - 32) \\C &= \frac{5}{9}(392 - 32) \\C &= \frac{5}{9}(360) \\C &= 200\end{aligned}$$

Step 2: Subtract the starting temperature.

$$200\text{ }^{\circ}\text{C} - 20\text{ }^{\circ}\text{C} = 180\text{ }^{\circ}\text{C}$$

Step 3: Determine the number of 15-minute cycles needed to heat the compound to its optimum temperature.

$$180 \div 9 = 20$$

Step 4: Multiply the number of cycles needed by 15 minutes and convert the product to hours and minutes.

$$\begin{aligned}15\text{ minutes} \times 9 &= 135\text{ minutes} \\135\text{ minutes} &= 2\text{ hours } 15\text{ minutes}\end{aligned}$$

So, it will take 2 hours and 15 minutes for the compound to reach its optimum temperature.

YOUR TURN 9.48

1. After reaching a temperature of 302 °F a chemical compound cools at the rate of 5 °C every 6 minutes. How long will it take until the compound has reached a temperature of zero degree Celsius.

VIDEO

[Learn the Metric System in 5 Minutes \(https://openstax.org/r/Metric_System_in_5_Minutes\)](https://openstax.org/r/Metric_System_in_5_Minutes)

Check Your Understanding

For the following exercises, determine the most reasonable value for each temperature.

34. Popsicle:
28.5 °C or

28.5 °F

35. Room temperature:

20 °F or

20 °C

36. Hot coffee:

71.1 °C or

71.1 °F

For the following exercises, use mental math to approximate the temperature.

37. $450\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$

38. $35\text{ }^{\circ}\text{C} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{F}$

39. $100\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$

For the following exercises, convert the temperatures to the nearest degree.

40. $225\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$

41. $27\text{ }^{\circ}\text{C} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{F}$

42. $750\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$



SECTION 9.5 EXERCISES

For the following exercises, determine the most reasonable value for each temperature.

1. Human body:

37 °C or

37 °F

2. An ice cube:

32 °C or

0 °C

3. Boiling water:

212 °C

or 212 °F

4. Summer day:

8 °C,

23 °C, or

75 °C

5. Winter day:

33 °C,

30 °C, or

3 °C

6. Lava:

700 °C,

70 °C, or

7 °C

For the following exercises, use mental math to approximate the temperature.

7. $500\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$

8. $25\text{ }^{\circ}\text{C} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{F}$

9. $350\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$

10. $150\text{ }^{\circ}\text{C} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{F}$

11. $-4\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$

12. $73\text{ }^{\circ}\text{C} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{F}$

13. $72\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$

14. $10\text{ }^{\circ}\text{C} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{F}$

15. $1,020\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$

For the following exercises, convert the temperatures to the nearest degree.

16. $450\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$

17. $35\text{ }^{\circ}\text{C} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{F}$

18. $525\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$

19. $140\text{ }^{\circ}\text{C} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{F}$

20. $-40\text{ }^{\circ}\text{F} = \underline{\hspace{1cm}}\text{ }^{\circ}\text{C}$

21. $67\text{ }^{\circ}\text{C} = \underline{\hspace{2cm}}\text{ }^{\circ}\text{F}$
22. $85\text{ }^{\circ}\text{F} = \underline{\hspace{2cm}}\text{ }^{\circ}\text{C}$
23. $15\text{ }^{\circ}\text{C} = \underline{\hspace{2cm}}\text{ }^{\circ}\text{F}$
24. $1,200\text{ }^{\circ}\text{F} = \underline{\hspace{2cm}}\text{ }^{\circ}\text{C}$
25. $112\text{ }^{\circ}\text{F} = \underline{\hspace{2cm}}\text{ }^{\circ}\text{C}$
26. $105\text{ }^{\circ}\text{C} = \underline{\hspace{2cm}}\text{ }^{\circ}\text{F}$
27. $125\text{ }^{\circ}\text{F} = \underline{\hspace{2cm}}\text{ }^{\circ}\text{C}$
28. A cup of tea is $30\text{ }^{\circ}\text{C}$. After adding ice, the temperature of the tea decreased $3\text{ }^{\circ}\text{C}$. What is the temperature of the tea now?
29. A liquid with a temperature of $15\text{ }^{\circ}\text{C}$ is placed on a stovetop. The liquid is heated at the rate of $1.5\text{ }^{\circ}\text{C}$ per minute. What is the temperature of the liquid after 10 minutes?
30. The temperature inside a store is $22\text{ }^{\circ}\text{C}$. Outside the temperature is $37\text{ }^{\circ}\text{C}$. How much cooler is it in the store than outside?
31. A boiling pan of water is $212\text{ }^{\circ}\text{F}$. As the water cools the temperature drops $6\text{ }^{\circ}\text{C}$ every 2 minutes. How many minutes until the temperature of the water reaches $40\text{ }^{\circ}\text{C}$?
32. On May 1 the temperature was $79\text{ }^{\circ}\text{F}$. On June 1 the temperature was $25\text{ }^{\circ}\text{C}$. Which day was warmer?
33. For each log placed on a fire, the temperature increases $45\text{ }^{\circ}\text{C}$. How many logs are needed for the campfire to increase $315\text{ }^{\circ}\text{C}$?
34. The temperature of a Bunsen burner flame increases $572\text{ }^{\circ}\text{F}$ each minute. About how many minutes does it take for the flame to increase $1,200\text{ }^{\circ}\text{C}$?
35. The instructions to cook a pizza say to set the oven at $425\text{ }^{\circ}\text{F}$. To the nearest degree, what is the temperature in degrees Celsius?
36. In the evening the temperature was $3\text{ }^{\circ}\text{C}$. By morning, the temperature had fallen $4\text{ }^{\circ}\text{C}$. What is the temperature now?
37. An 8,000 BTU air conditioner can cool a room $1\text{ }^{\circ}\text{C}$ every 5 minutes. If the temperature in the room is $24\text{ }^{\circ}\text{C}$, how long will it take the air conditioner to cool the room to $20\text{ }^{\circ}\text{C}$?
38. What is $95\text{ }^{\circ}\text{F}$ in degrees Celsius?
39. What is $50\text{ }^{\circ}\text{C}$ in degrees Fahrenheit?
40. Which is a reasonable value for the temperature of a room:
 - $60\text{ }^{\circ}\text{C}$,
 - $40\text{ }^{\circ}\text{C}$, or
 - $20\text{ }^{\circ}\text{C}$?

Chapter Summary

Key Terms

9.1 The Metric System

- metric system
- meter (m)
- gram (g)
- liter (L)
- square meter (m^2)
- cubic meter (m^3)
- degrees Celsius ($^{\circ}\text{C}$)
- conversion factor

9.2 Measuring Area

- area
- square units

9.3 Measuring Volume

- volume
- cubic units

9.4 Measuring Weight

- mass

9.5 Measuring Temperature

- temperature

Key Concepts

9.1 The Metric System

- The metric system is decimal system of weights and measures based on base units of meter, liter, and gram. The system was first proposed in 1670 and has since been adopted as the International System of Units and used in nearly every country in the world.
- Each successive unit on the metric scale is 10 times larger than the previous one. To convert between units with the same base unit, you must either multiply or divide by a power of 10.
- The most used prefixes are listed in [Table 9.1](#). An easy way to remember the order of the prefixes, from largest to smallest, is the mnemonic *King Henry Died from Drinking Chocolate Milk*.

9.2 Measuring Area

- Area describes the size of a two-dimensional surface. It is the amount of space contained within the lines of a two-dimensional space.
- Area is measured in square meters units; in the metric system the base unit for area is square meters (m^2).

9.3 Measuring Volume

- Volume is a measure of the space contained within or occupied by three-dimensional objects.
- Volume is measured in cubic units; the base unit for volume in the metric system is cubic meters (m^3)
- The liter (L) is a metric unit of capacity but is often used to express the volume of liquids. One liter is equivalent to one cubic decimeter, which is the volume of a cube measuring $10\text{ cm} \times 10\text{ cm} \times 10\text{ cm}$.

9.4 Measuring Weight

- Mass is the amount of matter in an object whereas weight is the force exerted on an object by gravity. The mass of an object never changes; the weight of an object changes depending on the force of gravity. An object with the same mass would weigh less on the moon than on Earth because the moon's gravity is less than that of Earth.
- In the metric system, weight and mass are often used interchangeably and are expressed in terms in grams or kilograms.

9.5 Measuring Temperature

- Temperature is a measure of how fast atoms and molecules are moving in a substance, whether that be the air, a stove top, or an ice cube. The faster those atoms and molecules move, the higher the temperature.
- In the metric system, temperature is measured using the Celsius (°C) scale.
- The Celsius scale was created with 100 degrees separating the point at which water freezes, 0 °C, and the point at which water boils, 100 °C. Scientifically, these are the points at which water molecules change from one state of matter to another—from solid (ice) to liquid (water) to gas (water vapor).

Videos

9.1 The Metric System

- [U.S. Office of Education: Metric Education \(https://openstax.org/r/U.S._Office_Education\)](https://openstax.org/r/U.S._Office_Education)
- [Neil deGrasse Tyson Explains the Metric System \(https://openstax.org/r/Tyson_Explains_Metric_System\)](https://openstax.org/r/Tyson_Explains_Metric_System)

9.2 Measuring Area

- [Converting Metric Units of Area \(https://openstax.org/r/Metric_Units_of_Area\)](https://openstax.org/r/Metric_Units_of_Area)
- [Why the Metric System Matters \(https://openstax.org/r/Metric_System_Matters\)](https://openstax.org/r/Metric_System_Matters)

9.3 Measuring Volume

- [How to Convert Cubic Centimeters to Cubic Meters \(https://openstax.org/r/Convert_CC_to_CM\)](https://openstax.org/r/Convert_CC_to_CM)
- [Converting Metric Units of Volume \(https://openstax.org/r/Converting_Metric_Units_of_Volume\)](https://openstax.org/r/Converting_Metric_Units_of_Volume)

9.4 Measuring Weight

- [Metric System: Units of Weight \(https://openstax.org/r/Metric_System:_units_of_weight\)](https://openstax.org/r/Metric_System:_units_of_weight)
- [Using Drones to Weigh Whales? \(https://openstax.org/r/Using_drones_to_weigh_whales?\)](https://openstax.org/r/Using_drones_to_weigh_whales?)
- [Metric Units of Mass: Convert mg, g, and kg \(https://openstax.org/r/Convert_mg-g-kg\)](https://openstax.org/r/Convert_mg-g-kg)

9.5 Measuring Temperature

- [Misconceptions About Temperature \(https://openstax.org/r/Misconceptions_About_Temperature\)](https://openstax.org/r/Misconceptions_About_Temperature)
- [Temperature Conversion Trick \(https://openstax.org/r/Temperature_Conversion_Trick\)](https://openstax.org/r/Temperature_Conversion_Trick)
- [Learn the Metric System in 5 Minutes \(https://openstax.org/r/Metric_System_in_5_Minutes\)](https://openstax.org/r/Metric_System_in_5_Minutes)

Formula Review

9.1 The Metric System

You can convert between unit sizes with the same base unit using the conversion factors shown in [Figure 9.4](#).

9.2 Measuring Area

To determine the area of rectangular-shaped objects:

$$\text{Area} = \text{length } (l) \times \text{width } (w)$$

or

$$A = l \times w$$

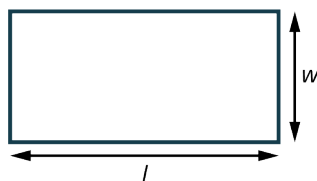


Figure 9.16 Rectangle with Length (l) and Width (w) Labeled

You can convert between metric area units using the conversion factors shown in [Figure 9.7](#).

9.3 Measuring Volume

To determine the volume of a rectangular prism:

$$\text{Volume} = \text{length } (l) \times \text{width } (w) \times \text{height } (h)$$

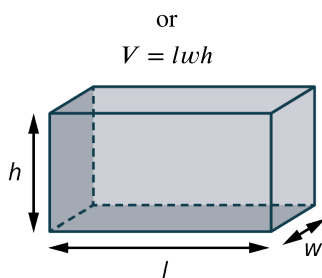


Figure 9.17 Rectangular Prism with Height (h), Length (l), and Width (w) Labeled

You can convert between metric volume units and metric capacity units using the relationships shown in [Table 9.2](#).

9.4 Measuring Weight

You can convert between metric weight units using the conversion factors shown in [Figure 9.13](#).

9.5 Measuring Temperature

To convert temperature from Fahrenheit to Celsius:

$$C = \frac{5}{9}(F - 32)$$

To convert temperature from Celsius to Fahrenheit:

$$F = \frac{9}{5}C + 32$$

To estimate temperature from Fahrenheit to Celsius:

$$C = \frac{F - 30}{2}$$

To estimate temperature from Celsius to Fahrenheit:

$$F = 2C + 30$$

Projects

Cooking

1. Take a favorite recipe that uses customary measures and convert the measures and cooking temperature to the metric system.
2. Find a recipe that uses metric measures and convert the measures and cooking temperature to the U.S. Customary System of Measurement, using cups, tablespoons, or teaspoons as required.
3. What did you observe? Was it easier to convert from one system to another? Which system allows for more precise measurements? What kitchen tools would you need in your kitchen if you used the metric system?

Shopping

1. Compare the average gas price in California to the average gas price in Puerto Rico.
2. What conversions did you need to make to do the comparison?
3. Do you think that the price of the gasoline is affected by the units in which it is sold?

Sports

1. What system of measurement is used for track and field events? Why do you think this system is used?
2. What system of measurement is used for football? Why do you think this system is used?
3. Research various sports records. Which units of measurement is used? What do you think influenced the unit of measure used?

Chapter Review

The Metric System

For the following exercises, determine the base unit of the metric system described: liter, gram, or meter.

1. amount of juice in a carton
2. weight of a puppy
3. height of a wall

For the following exercises, choose the smaller of the two units.

4. hectogram or centigram
5. decameter or kilometer
6. milliliter or deciliter
7. Convert 236 cm to meters (m).
8. Convert 137.5 mg to grams (g).
9. You purchase 15 kg of birdseed. You divide the birdseed into 10 bags. How many grams of birdseed are in each bag?
10. A track is 650 m long. If you run 8 laps around the track, how many kilometers did you run?

Measuring Area

For the following exercises, determine the most reasonable value for each area.

11. bedroom floor
 - 12 km²
 - 12 m²
 - 12 cm²
 - 12 mm²
12. nature preserve
 - 800 km²
 - 800 m²
 - 800 cm²
 - 800 mm²
13. desk surface
 - 0.125 km²
 - 0.125 m²
 - 0.125 cm²
 - 0.125 mm²

For the following exercises, convert the given area to the units shown.

14. 650,000 cm² = _____ m²
15. 17 m² = _____ cm²
16. 759 cm² = _____ mm²
17. A laptop is 30 cm by 20 cm. A sticker on the laptop cover measures 15 mm by 15 mm. How many square centimeters of the laptop cover is still visible?
18. A dining room wall is 2.75 m by 2.25 m. A window on the wall is 0.75 m by 1.5 m. How many square meters is the surface of the wall?
19. A floor is 6 m by 4 m. A rug on the floor measures 200 cm by 100 cm. How much of the floor, in m², is not covered by the rug?
20. A ceiling is 4 m by 2.5 m. A skylight in the ceiling measures 75 cm by 80 cm. How much of the ceiling, in m², would need to be painted if you decide to change the color of the ceiling?

Measuring Volume

For the following exercises, determine the most reasonable value for each volume.

21. aquarium
 - 65,750 km³
 - 65,750 m³
 - 65,750 cm³
 - 65,750 mm³

22. carton of soup
 94,000 L
 94 L
 940,000 mL
 940 mL
23. gift box
 6,250 km³
 6,250 m³
 6,250 cm³
 6,250 mm³

For the following exercises, convert the given volume to the units shown.

24. 59,837 mm³ = _____ cm³
25. 12.5 dm³ = _____ mL
26. 368 cm³ = _____ mm³
27. A box has dimensions of 15 cm × 10 cm × 25 cm. The box currently holds 1,750 cm³ of black beans. How many cubic centimeters of black beans are needed to completely fill the box?
28. The dimensions of a small storage unit are 3 m × 3 m × 3 m. What is the volume of a medium storage area with dimensions twice the size of the small unit?
29. How much liquid, in liters, can a container with dimensions of 35 cm × 15 cm × 20 cm hold?
30. A box is 20 cm long and 15 cm wide. The volume of the box is 2,100 cm³. What is the height of the box?

Measuring Weight

31. What is the most reasonable estimate for the weight of a small stack of coins?
 120 mg
 12 g
 1.2 kg
 0.12 mt

For the following exercises, convert the given weight to the units shown.

32. 7,900 g = _____ kg
33. 81 g = _____ mg
34. 0.12 kg = _____ g
- For the following exercises, determine the total weight in the units shown.
35. four 450 mg tablets _____ g
36. six 45 g candy bars _____ kg
37. seven 1.5 g buttons _____ mg
38. A box of chocolate weighs 2.75 kg. If you share the candy bar equally with 5 friends, how many grams of chocolate does each person get?
39. You purchase 15 tomatoes that weigh 100 grams each. If tomatoes cost \$1.18 per kilogram, how much did you pay?
40. A value-size package of ground chicken costs \$23.00. The package weighs 2.5 kg. What is the cost per gram of the chicken?

Measuring Temperature

For the following exercises, convert the temperatures to the nearest degree.

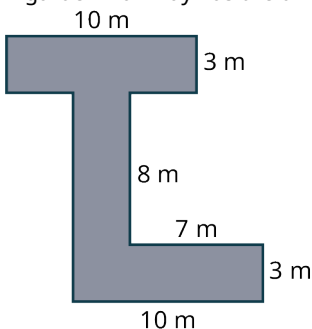
41. 365 °F = _____ °C
42. 42 °C = _____ °F
43. 575 °F = _____ °C
44. 135 °C = _____ °F
45. -20 °F = _____ °C
46. 73 °C = _____ °F
47. The temperature of a welder's torch increases 1,112 °F every 3 minutes. About how many minutes does it take for the torch to increase 1,200 °C?
48. A substance is heated to 302 °F. As the substance cools the temperature drops 4 °C every 3 minutes. How many

minutes until the temperature of the water reaches 130°C ?

49. On November 1 the temperature was 40°F . On December 1 the temperature was 7°C . Which day was cooler?
50. The instructions to cook lasagna say to set the oven at 475°F . To the nearest degree, what is the temperature in degrees Celsius?

Chapter Test

1. A student consumed 10 cans of their favorite soda last week. If each can contained 39 g of sugar, how many milligrams of sugar did they consume?
2. An athlete ran a total distance of 154 km in one month while training for a 5K fun run. How many meters did they run?
3. A doctor recommends their patient drinks 5 liters of water on days of vigorous exercise. If the patient drinks 3,400 mL, how much more do they need to drink?
4. A garden walkway has the dimensions shown. How many square meters of brick are needed to cover the walkway?



5. A studio apartment has a length of 1,200 cm and a width that one half of the length. What is the area of the apartment in meters?
6. A living room is 7 m long and 8 m wide. Carpet costs \$9.50 per square meter. How much will it cost to carpet the room?
7. Arif mixed two cartons of orange juice that each contain 600 cL, two bottles of cranberry juice that each contain 1,250 mL, and one 2-liter bottle of soda water to make punch for a party. How much punch, in liters, did Arif make?
8. A fish tank has the dimensions 40 cm $\times$ 20 cm $\times$ 25 cm. What is the capacity of the fish tank in liters?
9. A recipe for chili makes 5.4 L of chili. If a restaurant serves chili in 300 mL bowls, how many bowls of chili can they serve?
10. If a person weighs 76 kg but loses 3 kilograms, how much does the person weigh in grams?
11. A car weighs approximately 2 metric tons (mt). If a used car dealer has 222,000 kg of cars in their lot, how many cars are in the lot?
12. A restaurant has 126 kg of rice. If each serving contains 125 g, how many servings will they make?
13. A boiling pan of water is 212°F . As the water cools the temperature drops 12°C every 3 minutes. How many minutes until the temperature of the water reaches 28°C ?
14. On May 1 the temperature was 75°F . On June 1 the temperature was 28°C . Which day was warmer?
15. For each log placed on a fire, the temperature increases 45°C . How many logs are needed for the campfire to increase 315°C ?